



Qatar University

College of Engineering

**Department of Computer Science and
Engineering**

Project Report

***Secure Car Key Fob System Resistant to Replay and
Relay Attacks***

Project Group Members:

Aldana Khalid Altamimi (202109284)

Amna Abdulaziz Alobaidli (202005930)

Latifa Ali Albuennain (202103203)

Muneera Abdulla Alkubaisi (202105303)

Supervisor: Dr. Qutaibah Malluhi

Declaration

This report has not been submitted for any other degree at this or any other University. It is solely the work of us except where cited in the text or the Acknowledgements page. It describes work carried out by us for the capstone design project. We are aware of the university's policy on plagiarism and the associated penalties and we declare that this report is the product of our own work.

Student: Aldana AlTamimi

Date: 12/05/2026

Signature: *Aldana*

Student: Amna Alobaidi

Date: 12/05/2026

Signature: *Amna*

Student: Latifa

Date: 12/05/2026

Signature: *latifa*

Student: Muneera AlKubaisi

Date: 12/05/2026

Signature: *Muneera*

Abstract

This project presents the design, implementation, and evaluation of a Secure Wireless Vehicle Access System that addresses vulnerabilities in Remote Keyless Entry (RKE) systems. While such systems offer convenience, many are vulnerable to replay and relay attacks due to the use of static or predictable authentication mechanisms. This project aims to demonstrate these vulnerabilities and develop a practical security solution suitable for embedded environments.

The system follows a three-stage approach. In Stage 1, a baseline system is implemented using static RF communication to perform lock and unlock operations, with LED and buzzer feedback indicating system states. In Stage 2, system vulnerabilities are demonstrated through replay and relay attacks, where an attacker device captures and retransmits signals to gain unauthorized access, highlighting the weaknesses of static communication.

To address these issues, Stage 3 introduces a rolling code-based authentication mechanism. Each transmission incorporates a dynamically changing counter combined with a non-linear polynomial transformation and XOR operation using a shared secret key. This ensures that every transmitted signal is unique and prevents reuse of previously captured signals. The receiver independently verifies incoming messages while enforcing replay protection and synchronization to maintain reliability.

The system integrates embedded hardware and software components and was validated through comprehensive testing. Results confirm that the baseline system is vulnerable, while the rolling code mechanism effectively mitigates replay attacks and improves overall system security with minimal impact on performance.

The main contribution of this work is a low-cost and modular platform that demonstrates both wireless security vulnerabilities and practical countermeasures within a single system. The structured design provides a clear transition from insecure to secure communication, offering both educational value and a foundation for further development in secure embedded systems.

Acknowledgment

We would like to express our sincere gratitude to **Dr. Qutaibah** for his continuous guidance, valuable feedback, and support throughout the duration of this project. His supervision played a crucial role in shaping the direction and quality of our work.

We also extend our appreciation to **Engineer Tamer** for his assistance during the development and testing phases, and for providing practical insights that helped improve the system implementation.

Finally, we would like to thank everyone who contributed to our learning experience and supported us during this project.

Table of Contents

1. Introduction and Motivation	1
1.1 Problem Statement	2
1.2 Project Objectives	3
1.3 Project Significance and Expected Benefits	4
1.4 Impact Analysis (Global, Economic, Environmental, Social)	6
1.5 Market Research and Business Viability	9
1.6 Complex Engineering Problem Justification	12
2. Background and Related Work	14
2.1 Background	14
2.2 Related Work	16
3. Requirements Analysis	18
3.1 Functional Requirements	19
3.2 Design Constraints	23
3.3 Design Standards	26
3.4 Professional Code of Ethics	27
3.5 Assumptions	29
4. Project Plan	30
4.1 Project Milestones	3
4.2 Project Timeline	31
4.3 Risk Analysis	32
5. Solution Design	33
5.1 Design Overview	33
5.2 High-Level Architecture	34
5.3 Design Alternatives	36
5.4 Hardware/Software Used	38
5.5 Hardware Design	39
5.6 Software Design	42
6. Implementation	45
6.1 System Implementation Overview	45
6.2 Hardware Implementation	46
6.3 Software Implementation	47
6.4 Challenges and Solutions	4
7. Testing and Evaluation	50
7.1 Testing Overview	50
7.2 Stage 1 Testing	51
7.3 Stage 2 Testing	52
7.4 Stage 3 Security Testing	53
7.5 Results and Analysis	55
8. Analysis of the Impact of the Engineered Solution	57

9. **Conclusion** 59

10. **Future Work** 62

11. **References** 64

12. **Appendix** 66

List of Figures

Figure	Description	Page
Figure 1	Annual Number of Registered Vehicles in Qatar (2011–2022).....	16
Figure 2	High-Level Architecture of the Proposed System.....	20
Figure 3	Stage 1 Static Communication Workflow.....	22
Figure 4	Replay Attack Scenario.....	23
Figure 5	Relay Attack System Using Dual Devices.....	24
Figure 6	Rolling Code Authentication Workflow (Stage 3)	25
Figure 7	Remote Key Unit Hardware Design.....	28
Figure 8	Vehicle Unit Hardware Design.....	29
Figure 9	Replay Attack Hardware Subsystem.....	30
Figure 10	Secure Rolling Code Hardware Design.....	31
Figure 11	Output Feedback Subsystem.....	32
Figure 12	Software Architecture Diagram.....	34
Figure 13	Key Software Flowchart.....	35
Figure 14	Vehicle Software Flowchart.....	36
Figure 15	Rolling Code Sequence Diagram.....	37

List of Tables

Table	Description	Page
Table 1	Expected Benefits and Impacts Across Different Contexts.....	15
Table 2	Summary of Related Work.....	18
Table 3	Design Constraints.....	27
Table 4	Design Standards Used in the Project.....	27
Table 5	Ethical Issues and Responsibilities.....	28
Table 6	Project Milestones.....	29
Table 7	Project Timeline.....	29
Table 8	Potential Risks and Mitigation Strategies.....	31
Table 9	Comparison of Design Alternatives.....	33
Table 10	Hardware and Software Components.....	35
Table 11	Summary of Hardware Design Choices.....	36
Table 12	Rolling Code Signal Format.....	37
Table 13	Verification of System Design Constraints.....	41
Table 14	Impact Analysis Across Different Contexts.....	43

1. Introduction and Motivation

Wireless Remote Keyless Entry (RKE) systems have become a standard feature in modern vehicles, providing users with convenient and seamless access without the need for traditional mechanical keys. By enabling remote locking and unlocking through radio frequency (RF) communication, these systems significantly improve usability and user experience. However, this convenience introduces critical security concerns, as wireless signals can be intercepted and manipulated by unauthorized parties [1].

One of the primary challenges in RKE systems is their vulnerability to signal-based attacks. Many low-cost and legacy systems rely on static or weakly protected communication protocols, making them susceptible to attacks such as replay and relay attacks [1], [2]. In a replay attack, an adversary captures a valid transmission and reuses it later to gain unauthorized access. In a relay attack, attackers extend the communication range between the key and the vehicle in real time, deceiving the system into granting access even when the legitimate key is physically distant. These vulnerabilities pose serious risks, including unauthorized vehicle access, theft, and reduced trust in wireless technologies.

Securing RKE systems is particularly challenging due to the constraints of embedded environments. Such systems must balance security with limited computational resources, low power consumption, real-time responsiveness, and cost efficiency [3]. Implementing robust security mechanisms under these constraints requires careful engineering trade-offs, making the problem both technically complex and practically important.

This project addresses these challenges through a structured three-stage approach. First, a baseline system is developed using static RF communication to replicate real-world vulnerabilities. Second, replay and relay attacks are implemented to demonstrate how such systems can be exploited. Finally, a secure rolling code-based authentication mechanism is introduced to replace static communication. This mechanism uses a counter-based approach combined with a non-linear

transformation and a shared secret key to ensure that each transmitted signal is unique and resistant to reuse. Through this progression, the project highlights existing security weaknesses and demonstrates a practical and reliable approach to improving wireless system security in embedded environments.

1.1 Problem statement

Wireless Remote Keyless Entry (RKE) systems are widely used for vehicle access due to their convenience and ease of use. However, many existing implementations—particularly in older vehicles and low-cost systems—lack robust security mechanisms. These systems often rely on static or weakly protected RF communication, making them vulnerable to signal interception and unauthorized access [1], [2]. As a result, attackers can exploit these weaknesses using techniques such as replay and relay attacks, leading to serious risks such as unauthorized vehicle access and reduced user trust in wireless technologies.

The core problem addressed in this project is the lack of a secure, practical, and low-cost solution that both demonstrates these vulnerabilities and provides an effective countermeasure suitable for embedded systems. There is a need for a system that not only replicates real-world weaknesses in wireless keyless access but also implements a reliable and efficient security mechanism that can operate under resource constraints [3]. The project aims to demonstrate modern security techniques through a clear, low-cost, and educational prototype.

Technical Challenges

- Designing reliable RF communication between transmitter and receiver under real-world conditions (noise, interference, and signal attenuation).
- Implementing replay and relay attack mechanisms in a controlled and accurate manner.
- Developing a lightweight and efficient authentication mechanism (e.g., rolling code) suitable for low-power embedded systems.

- Ensuring synchronization between the key and vehicle in counter-based authentication systems.
- Managing latency and timing constraints in real-time communication.
- Balancing system performance with limited hardware resources such as processing power, memory, and energy consumption.

Non-Technical Challenges

- Maintaining system simplicity and usability while introducing security mechanisms.
- Ensuring ethical execution of attack demonstrations within controlled environments.
- Managing project timeline, teamwork coordination, and task distribution.
- Keeping the system cost-effective and suitable for educational and practical deployment.
- Presenting complex security concepts in a clear and understandable manner.

This problem is significant because it reflects real-world vulnerabilities in widely deployed systems. Addressing it requires not only technical implementation but also careful consideration of usability, cost, and reliability, making it a comprehensive engineering challenge.

1.2 Project objectives

The primary objective of this project is to design, implement, and evaluate a secure wireless vehicle access system that demonstrates existing vulnerabilities and provides practical countermeasures. The objectives are defined in measurable terms to enable clear evaluation of system performance and success.

Objective 1: Baseline System and Attack Demonstration

Design and implement a functional wireless keyless entry system using static RF communication, then demonstrate replay and relay attacks to evaluate system vulnerabilities.

Desired Results:

- Reliable lock/unlock communication with $\geq 95\%$ success rate
- Successful replay and relay attack demonstrations under controlled conditions

Objective 2: Secure Rolling Code Implementation

Develop and implement a rolling code-based authentication mechanism using a synchronized counter, non-linear transformation, and shared secret key.

Desired Results:

- Generate unique encoded signals for each transmission
- Prevent reuse of previously captured signals
- Improve resistance against unauthorized access attempts

Objective 3: System Evaluation and Validation

Test and compare system behavior before and after applying the security mechanism while evaluating reliability, latency, and operational stability.

Desired Results:

- Replay attacks rejected in secure mode
- Stable communication and low-latency performance
- Consistent operation during repeated testing

Objective 4: Documentation and Prototype Demonstration

Produce a complete technical report and a functional prototype demonstrating all project stages.

Desired Deliverables:

- Fully functional hardware prototype
- Demonstration of vulnerabilities and security improvements
- Comprehensive report covering design, implementation, and evaluation

These objectives collectively ensure that the project can be evaluated based on functionality, security effectiveness, system performance, and clarity of documentation.

1.3 Project significance and expected benefits

The increasing reliance on wireless technologies in modern vehicles has made Remote Keyless Entry (RKE) systems a critical component of everyday security. While these systems offer significant convenience, many implementations—especially in older or low-cost vehicles—remain vulnerable to replay and relay attacks due to weak or static authentication mechanisms. Such vulnerabilities expose users to risks including unauthorized access, vehicle theft, and privacy breaches. Without effective countermeasures, the reliability and trust in wireless automotive systems continue to decline.

Significance of the Project

This project addresses a real-world cybersecurity problem that affects a large number of users. Research in vehicular security has shown that insecure RKE systems can be exploited using inexpensive and widely available hardware, making such attacks both practical and widespread. By identifying and addressing these vulnerabilities, the project contributes to improving user safety, protecting assets, and enhancing confidence in wireless technologies.

If these issues remain unresolved, the continued use of insecure systems may lead to increased incidents of unauthorized access and financial loss, as well as reduced trust in modern vehicle technologies.

Contribution of the Work

The main contribution of this project lies in bridging the gap between theoretical cybersecurity concepts and practical embedded system implementation. Unlike purely theoretical studies, this work provides a functional prototype that:

- Demonstrates real-world vulnerabilities (replay and relay attacks)

- Implements a practical security mechanism using a rolling code-based approach
- Enables direct comparison between insecure and secure system behavior

This approach enhances academic understanding by offering a hands-on platform for learning and experimentation in embedded systems and cybersecurity. It also promotes the principle of security by design, encouraging early integration of protection mechanisms in system development.

Non-Triviality and Innovation

This project is non-trivial because the problem cannot be solved by directly applying existing commercial solutions, which are often proprietary and inaccessible for academic use. Instead, a custom solution was designed and implemented from the ground up.

The complexity of the project arises from:

- Integrating hardware and software subsystems, including RF communication and embedded processing
- Implementing secure authentication within resource-constrained environments
- Maintaining synchronization and reliability in wireless communication

The innovative aspect of this project lies in its structured three-stage framework, where the system:

1. Replicates real-world vulnerabilities
2. Demonstrates practical attack scenarios
3. Implements and evaluates an effective security mechanism

The use of a lightweight rolling code approach provides a practical balance between security and performance, making it suitable for low-cost embedded systems.

Personal Motivation and Career Relevance

The motivation behind this project is driven by an interest in embedded systems, wireless communication, and cybersecurity. It provides an opportunity to apply

theoretical knowledge in a practical setting, particularly in designing and implementing secure communication mechanisms.

From a career perspective, the project supports the development of key skills in:

- Secure embedded system design
- Wireless communication protocols
- Cybersecurity analysis and threat evaluation

These skills are highly relevant to fields such as automotive security, Internet of Things (IoT) systems, and smart device development. By addressing a real-world problem with practical impact, the project helps build a strong foundation for future work in secure system design and advanced technologies.

1.4 Analysis of global, economic, environmental, and social impact

The proposed secure wireless vehicle access system extends beyond a purely technical solution by contributing to broader global, economic, environmental, and societal contexts. By improving the security of widely used wireless systems, the project addresses real-world challenges related to safety, sustainability, and trust in technology.

Global Impact

Wireless security is a global concern, particularly with the rapid growth of connected devices and smart transportation systems. Vulnerabilities in RKE systems affect users worldwide, especially in regions where older or low-cost systems are still in use. This project contributes to global efforts in enhancing cybersecurity awareness by demonstrating both vulnerabilities and practical countermeasures. It supports the development of more secure wireless systems aligned with modern trends in IoT and smart mobility, ultimately contributing to improved safety and reduced risks associated with unauthorized access.

Social Impact

The project has a direct positive impact on society by improving personal safety and reducing the likelihood of vehicle theft. A more secure system increases user

confidence in wireless technologies and reduces concerns related to security breaches. Additionally, the project raises awareness of cybersecurity risks among students and engineers, promoting responsible system design and encouraging a security-focused mindset. This contributes to building safer communities and more informed technology users.

Environmental Impact

From an environmental perspective, the system promotes sustainability through the use of low-power and energy-efficient components such as microcontrollers, LEDs, and buzzers. The replacement of mechanical components with electronic indicators reduces energy consumption and mechanical wear, extending system lifespan. Furthermore, improved vehicle security can indirectly reduce theft-related losses and the need for replacement components, contributing to lower resource consumption and reduced electronic waste. The project demonstrates that secure system design can be achieved while maintaining environmental responsibility.

Economic Impact

The proposed solution provides economic benefits at both individual and broader levels. For individuals, enhanced vehicle security reduces financial losses associated with theft, damage, and insurance costs. At a wider scale, reducing such incidents can lower the burden on law enforcement and insurance sectors.

From a production standpoint, the system is designed using low-cost and widely available components, making it suitable for scalable implementation and potential commercialization. Its affordability also makes it applicable for upgrading existing systems in older vehicles. This creates opportunities in the embedded systems and automotive accessory markets, contributing to innovation and economic growth in secure and smart mobility solutions.

Table 1. Expected benefits and impacts of various contexts

Context	Expected Benefits and Impacts
Public Health, Safety, and Welfare	Enhances user safety by reducing the risk of unauthorized vehicle access and theft. Improves overall trust in wireless security systems and contributes to safer daily usage of vehicles.
Global	Addresses a widespread cybersecurity issue affecting wireless systems globally. Promotes awareness of vulnerabilities in RKE systems and supports the development of more secure IoT and smart mobility technologies.
Economic	Reduces financial losses related to vehicle theft and damage. Supports low-cost implementation and potential commercialization, creating opportunities in embedded systems and automotive security markets.
Environmental	Utilizes energy-efficient electronic components, reducing power consumption. Minimizes reliance on mechanical parts, leading to lower wear and reduced electronic waste over time.
Societal	Increases public awareness of cybersecurity risks and encourages secure system design practices. Improves user confidence in wireless technologies and contributes to safer and more informed communities.

1.5 Market Research and Business Viability

The increasing adoption of wireless access systems in vehicles has created a growing demand for secure and affordable Remote Keyless Entry (RKE) solutions. Despite

technological advancements, a significant number of vehicles—particularly older models and aftermarket installations—still rely on insecure communication protocols, making them vulnerable to replay and relay attacks. This creates a clear market need for retrofit solutions that enhance security without requiring full system replacement.

Market Need and Market Size

Many existing RKE systems prioritize convenience over security, leaving users exposed to increasingly accessible attack techniques. This is especially relevant in regions with a high number of older vehicles or widespread use of low-cost RF-based systems.

In Qatar, the number of registered vehicles has reached approximately 1.7–1.8 million. It is estimated that around 30–40% of these vehicles may still use outdated or vulnerable keyless entry systems, representing a potential addressable market of 500,000 to 700,000 vehicles locally.

At a global level, the automotive security and IoT-based access control markets are expanding rapidly, increasing the scalability and relevance of the proposed solution

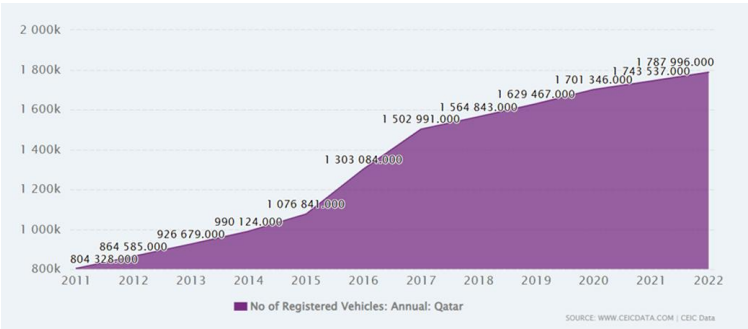


Figure 1. Annual Number of Registered Vehicles in Qatar (2011–2022)

Target Customers and Demographics

The primary target customers include:

- Individual Vehicle Owners: Owners of older vehicles or vehicles equipped with aftermarket keyless systems, typically aged 25–60 with moderate to high security awareness.
- Automotive Workshops and Installers: Local garages and car accessory shops that can act as distributors and installation providers.
- Fleet Operators and Commercial Users: Taxi companies, delivery services, and rental agencies that require enhanced security for multiple vehicles.

Competing Products and Differentiation

Existing market solutions include:

- Basic RF key fob systems using static or simple rolling codes
- Manufacturer-integrated RKE systems with proprietary security
- Advanced smart key systems with encrypted communication

However, these solutions have limitations:

- Low-cost systems often lack adequate security
- Commercial systems are closed and not customizable
- Advanced systems are expensive and not suitable for retrofitting

The proposed system differentiates itself by:

- Providing an open and customizable design
- Supporting both insecure and secure modes for analysis and testing
- Offering a low-cost retrofit solution with improved security
- Enabling real-time demonstration of vulnerabilities and countermeasures

Novel Features of the Product

- Three-stage architecture (Baseline → Attack → Secure System)
- Integration of replay and relay attack demonstrations
- Rolling code–based authentication mechanism
- Low-cost implementation using widely available components
- Dual-purpose design (educational and practical use)
- Synchronization mechanism to maintain reliability

These features make the system unique in both academic and practical contexts.

Pricing Strategy

The system is designed as a low-cost solution while providing enhanced security features. Based on component costs (microcontroller, RF modules, LEDs, buzzer, and supporting circuitry), the estimated price range is: QAR 110 – QAR 180

This positions the product above basic RF systems but below advanced commercial solutions, offering a competitive balance between cost and functionality.

Market Entry and Competitiveness Plan

To bring the product to market and ensure competitiveness, the following steps are required:

- Conduct further testing and validation to ensure reliability and security
- Develop a compact and user-friendly hardware enclosure
- Ensure compliance with RF communication and safety standards
- Collaborate with local automotive workshops for installation and distribution
- Develop targeted marketing strategies focusing on security awareness
- Maintain competitive pricing while emphasizing security advantages

The proposed solution addresses a clear market need by offering a cost-effective, secure, and scalable retrofit system. Its combination of practical functionality, educational value, and affordability positions it as a competitive solution in both local and global markets.

1.6 Justification of the problem as a complex engineering problem

The design and implementation of a secure Remote Keyless Entry (RKE) system resistant to replay and relay attacks represents a complex engineering problem due to the presence of conflicting requirements, multiple interacting subsystems, and real-world constraints. The problem extends beyond basic functionality and requires careful engineering judgment to balance security, performance, cost, and usability.

Conflicting Technical Requirements

A key source of complexity is the need to satisfy competing design requirements. The system must provide effective security mechanisms while maintaining:

- Low power consumption for battery-operated key devices
- Low cost to ensure affordability and scalability
- Fast response time to preserve user convenience

Improving security typically increases computational complexity and energy consumption, while simplifying the system reduces protection against attacks. In this project, an initial challenge-response approach was evaluated but found to be impractical due to RF communication limitations such as latency and packet loss. As a result, a rolling code-based mechanism was adopted as a balanced solution. This trade-off highlights the absence of a single optimal solution and the need for informed engineering decisions.

Multiple Components and Subsystems

The project consists of several interconnected subsystems, each addressing a different aspect of the problem:

- Key fob subsystem: Handles user input and RF signal transmission
- Car unit subsystem: Receives signals, processes commands, and controls outputs
- Attack modules: Implement replay and relay attack mechanisms
- Secure communication module: Implements authentication protocols (rolling codes)

Each subsystem must operate correctly both independently and as part of an integrated system. Synchronization, timing, and communication between these components introduce additional complexity, particularly in wireless environments where signal reliability is not guaranteed.

Multidisciplinary Nature

The system consists of several interconnected subsystems, each addressing a different aspect of the problem:

- Key fob subsystem: Handles user input and transmits RF signals

- Car unit subsystem: Receives signals, performs validation, and controls outputs
- Attack modules: Implement replay and relay attack scenarios
- Secure communication module: Implements rolling code-based authentication

These subsystems must function correctly both independently and as part of an integrated system. Synchronization, timing, and reliable communication between components introduce additional complexity, particularly in wireless environments.

Lack of a Single Obvious Solution

The problem does not have a straightforward or universally accepted solution. Existing systems use different approaches such as static codes, rolling codes, or proprietary cryptographic protocols, each with limitations. Many commercial solutions are closed and cannot be directly adapted for experimental use.

Therefore, this project required developing a custom solution that:

- Demonstrates real-world vulnerabilities
- Simulates practical attack scenarios
- Implements and evaluates an effective countermeasure

This open-ended nature further reinforces the complexity of the problem

Real-World Impact and Stakeholder Considerations

The project addresses a real-world issue with implications for multiple stakeholders:

- Users: Require secure, reliable, and easy-to-use systems
- Manufacturers: Need cost-effective and scalable solutions
- Researchers and educators: Require transparent and testable systems

Balancing these needs introduces additional constraints, as improvements in one area may affect another.

Technological and Societal Implications

The proposed solution contributes to improving cybersecurity awareness, reducing vehicle-related risks, and promoting secure design practices in embedded systems.

It must also be evaluated in terms of economic feasibility, ethical considerations (such as controlled attack demonstrations), and environmental impact.

Conclusion

This project qualifies as a complex engineering problem because it involves conflicting requirements, multiple interacting subsystems, multidisciplinary integration, and no single obvious solution, while addressing a real-world issue with significant impact. Solving this problem requires systematic design, critical analysis, and engineering trade-offs, making it representative of real-world engineering challenges.

2. Background and related work

2.1 Background

Remote Keyless Entry (RKE) systems have become a standard feature in modern vehicles due to the convenience they offer in locking, unlocking, and starting the car without the need for physical keys. These systems operate by having a handheld transmitter, or key fob, sending a radio-frequency signal to a receiver module inside the vehicle. Once the signal is verified, the vehicle performs the corresponding action, such as unlocking the doors.

Despite their popularity, many existing RKE systems, particularly those found in older vehicles or aftermarket installations, remain vulnerable to attacks. Traditional fixed-code implementations, which transmit the same signal every time, can easily be compromised through replay attacks, where an adversary captures the signal and replays it later to gain unauthorized access. Even systems that use rolling codes, which change with each transmission, have been shown to suffer more advanced attacks such as RollJam, where synchronization between the transmitter and receiver can be manipulated.

Survey studies on vehicular security emphasize that while rolling code schemes and challenge–response authentication protocols have improved resistance to simple replay attacks, they are still subject to significant challenges [1], [2]. One major issue is the limited computational power of legacy automotive microcontrollers, which restricts the use of modern cryptographic algorithms. Another concern is resynchronization: when signals are lost or blocked, keeping both devices aligned without introducing new vulnerabilities is complex. Furthermore, practical considerations such as cost, compatibility with older hardware, and user acceptance make the design of secure yet lightweight solutions even more difficult.

These surveys highlight that the evolution of RKE security has moved from fixed codes toward more advanced but still imperfect mechanisms. The continued presence of vulnerabilities in legacy systems underscores the need for new approaches that are secure, efficient, and cost-effective, which provides the foundation and motivation for this project.

2.2 Related work

The security of Remote Keyless Entry (RKE) systems has been an active area of research for more than a decade, as these systems are vulnerable to wireless signal interception and manipulation. Early studies by Francillon et al. [1] and Verdult et al. [2] demonstrated that many commercial key fob implementations use static or weak rolling codes, allowing attackers to easily perform replay and rolljam attacks. Their experiments showed that attackers could intercept a legitimate transmission and reuse it to unlock the car without the owner’s knowledge.

To counteract these vulnerabilities, researchers have proposed cryptographic protocols such as rolling code synchronization and challenge–response authentication. Garcia et al. [3]

presented a comprehensive survey on vehicular security mechanisms and discussed the weaknesses in proprietary algorithms like KeeLoq, which is still widely used in

low-cost key fobs. Similarly, Van Capelleveen [4] analyzed aftermarket RKE systems and highlighted that many lack strong encryption or mutual authentication, making them susceptible to relay-based attacks.

More recent studies focus on distance-bounding protocols to prevent relay attacks by measuring the signal propagation delay between the key and the vehicle [5]. Rasmussen and Capkun [6] introduced a prototype that limits the acceptable signal round-trip time to detect relayed signals, effectively mitigating distance extension attempts. Other approaches use cryptographic challenge-response methods implemented with AES or ECC algorithms to ensure that only a legitimate key can respond correctly within the allowed time window [7].

In addition, practical demonstrations have been conducted using low-cost Software Defined Radios (SDRs) to test the resilience of commercial systems. For example, researchers from Hamad Bin Khalifa University (HBKU) and Qatar University investigated the vulnerability of RKE protocols to jam-listen-replay attacks under local conditions, confirming that replay attacks can be carried out with inexpensive hardware [8]. Their work reinforces the importance of adopting strong synchronization and encryption mechanisms in modern car locking systems.

Finally, commercial solutions like rolling code generators and secure microcontrollers (e.g., NXP, Microchip, TI) are being integrated into modern vehicles to provide cryptographic security. However, these solutions can still be bypassed if the implementation lacks proper key management or synchronization recovery mechanisms [9][10]. Additional studies have highlighted further challenges: Shwartz et al. [11] demonstrated practical weaknesses in real-world keyless entry deployments; Kleanthous et al. [12] proposed lightweight authentication protocols tailored for embedded microcontrollers; Yildiz et al. [13] analyzed vulnerabilities in aftermarket RF remotes similar to those used in educational prototypes; Fernandes et al. [14] introduced behavioral-based keyfob authentication as an additional defense layer; and Lighari et al. [15] proposed improved rolling-code generation schemes that enhance synchronization and replay resistance.

More recent research continues to expand the RKE security landscape. Hines et al. [16] provided a modern evaluation of telematics-linked keyless entry vulnerabilities, showing how networked vehicle systems increase attack surfaces. Spreitzer et al. [17] documented how side-channel leakages can compromise cryptographic implementations in RKE systems, even when protocols are mathematically secure. Checkoway et al. [18] revealed that integrated automotive subsystems can be compromised through wireless entry points, emphasizing systemic risk beyond just the keyfob. Koscher et al. [19] further demonstrated that multi-vector attacks targeting both RF and in-vehicle networks can bypass security controls when wireless authentication is weak.

Therefore, further research such as the present project is essential to develop a practical, secure, and affordable RKE prototype that can withstand modern replay and relay attack techniques while remaining feasible for low-cost hardware implementations.

Table2: summary of Related work

Study	Focus / Contribution	Main Findings	Limitations
Francillon et al. [1]	Analysis of early RKE implementations using static or weak rolling codes	Demonstrated practical replay and rolljam attacks	Did not propose mitigation strategies
Verdult et al. [2]	Reverse-engineering commercial RKE algorithms	Revealed weak cryptography and protocol flaws	Focused mainly on older systems
Garcia et al. [3]	Survey of vehicular security mechanisms	Identified weak keys and lack of mutual authentication	Covers mainly low-cost systems

Van Capelleveen et al. [4]	Security evaluation of aftermarket RKE systems	Found missing encryption and relay vulnerabilities	Limited to selected aftermarket devices
Rasmussen & Capkun [5]	Distance bounding protocol for relay-attack prevention	Detects abnormal signal delays	Requires precise hardware timing
AES/ECC-based approaches [6]	Modern cryptographic challenge-response methods	Strong authentication prevents replay/relay attacks	Higher computational cost
Schwartz et al. [7]	Real-world keyless entry weakness analysis	Demonstrated practical deployment vulnerabilities	Limited mitigation discussion
Koushanous et al. [8]	Lightweight authentication for embedded controllers	Suitable for low-power devices	Prototype-level validation
Yildiz et al. [9]	RF remote replay-risk evaluation	Similar vulnerabilities in aftermarket remotes	Limited modern vehicle testing
Fernandes et al. [10]	Behavioral-based keyfob authentication	Additional defense layer using user behavior	Can affect usability
Uguri et al. [11]	Improved rolling-code generation schemes	Better synchronization	Increased implementation complexity

		and replay resistance	
Hines et al. [12]	Networked vehicle telematics vulnerabilities	Connected systems enlarge attack surface	Focused on telematics systems
Spreitzer et al. [13]	Side-channel leakage in crypto devices	Secret keys may leak from hardware	Requires close access / equipment
Checkoway et al. [14]	Integrated automotive subsystem compromise	Wireless interfaces can expose core systems	Broad attack model
Koscher et al. [15]	Multi-vector attacks on in-vehicle networks	Weak authentication can be bypassed	High technical complexity
Microchip secure MCU studies [16]	Secure MCU integration for keyless systems	Hardware-backed key storage improves security	Higher component cost
NXP automotive security studies [17]	Automotive secure controller solutions	Strong cryptographic support for RKE	Added design complexity
TI low-power secure systems [18]	Low-power secure wireless controller designs	Feasible for battery-powered key fobs	Performance trade-offs
Recent rolling-code research [19]	Synchronization recovery and resilience methods	Improved robustness against desync attacks	Requires protocol updates

3. Requirements analysis

This section defines the high-level requirements that the proposed system must satisfy based on the principles of engineering design. The objective is to develop a solution that fulfills functional requirements while operating within practical constraints such as cost, performance, reliability, usability, and security.

Engineering design in this project follows an iterative process, where system requirements are identified, analyzed, and refined through testing and evaluation. Multiple design alternatives were considered, particularly in selecting an appropriate security mechanism. An initial approach based on bidirectional authentication was evaluated but found to be impractical under real-world RF communication constraints. Consequently, a rolling code-based approach was adopted as a balanced solution that improves security while maintaining system reliability and real-time performance.

At a high level, the system requires the integration of several core components:

- Input Interface: Detects user commands (lock/unlock) from the key unit
- Wireless Communication Module: Enables RF signal transmission between key and vehicle
- Processing Unit: Performs encoding, decoding, and authentication operations
- Security Mechanism: Ensures that transmitted signals are unique and resistant to reuse
- Output Interface: Provides user feedback through LEDs and buzzer

These components collectively enable the system to perform secure wireless communication while maintaining efficiency and usability. The requirements defined in this section form the foundation for the subsequent design, implementation, and evaluation of the system.

3.1 Functional requirements

This section defines the functional requirements of the proposed secure wireless vehicle access system. These requirements describe the operations the system

must perform, the data it processes, and how users interact with it. The objective is to clearly define system behavior without requiring detailed technical knowledge.

A. Data and Signals Entered into the System

The system processes the following inputs:

- User Commands: Signals generated when the user initiates lock or unlock actions through the key unit.
- Wireless Signals: RF signals transmitted between the key and vehicle representing control commands.
- Captured Signals (Attack Mode): Signals intercepted and stored by the attacker device for replay purposes.
- Authentication Data (Secure Mode): Dynamic data including counter values and encoded results used for verification.

B. Operations and Processes

The system performs a series of operations depending on its operational stage:

1. Basic Operation (Stage 1 – Baseline System)

- Detect user input commands
- Generate corresponding RF signals
- Transmit signals to the receiver unit
- Receive and decode incoming signals
- Trigger system response based on recognized commands

2. Attack Simulation (Stage 2)

- Intercept wireless signals during transmission
- Store captured signals for later use
- Retransmit stored signals (replay attack)
- Forward signals in real time between devices (relay attack)

3. Secure Operation (Stage 3)

- Increment an internal counter on each transmission
- Apply a transformation function to the counter value
- Generate an encoded authentication value using a shared secret key

- Transmit the encoded message
- Recompute and verify the expected value at the receiver
- Accept or reject the command based on validation results

C. System Workflows

The system supports multiple workflows representing different operational scenarios:

Normal Operation Workflow

1. User initiates a command
2. System transmits a wireless signal
3. Receiver processes the signal
4. System performs the corresponding action

Replay Attack Workflow

1. Attacker captures a valid signal
2. Attacker stores the signal
3. Attacker retransmits the signal later
4. System executes the action without verifying authenticity

Relay Attack Workflow

1. Attacker intercepts a signal near the key
2. Signal is forwarded in real time to another device near the vehicle
3. Receiver accepts the signal as legitimate
4. Unauthorized access is granted

Secure Rolling Code Workflow

1. User initiates a command
2. Counter value is incremented
3. Encoded result is generated using transformation and secret key
4. Message is transmitted
5. Receiver computes expected value
6. Access is granted only if the values match and counter is valid

D. System Outputs and Actuators

The system produces the following outputs:

- State Indication Outputs: Visual and auditory feedback (LEDs and buzzer) indicating system status
- Wireless Transmission Outputs: RF signals representing commands and encoded data
- Diagnostic Outputs: Serial messages for monitoring and debugging
- Security Decision Outputs: Indication of accepted or rejected authentication

E. Main Users of the System

The system involves the following user roles:

- Primary User: Operates the system for lock and unlock actions
- Attacker (Simulation Role): Demonstrates vulnerabilities through replay and relay attacks in a controlled environment
- System Evaluator / Engineer: Tests and validates system functionality and security

F. Project Scope and Use Cases

The implementation focuses on the following key use cases:

- Use Case 1: Lock Operation The system processes a user command to secure the vehicle
- Use Case 2: Unlock Operation The system processes a user command to allow access
- Use Case 3: Wireless Communication The system demonstrates reliable RF signal transmission and reception
- Use Case 4: Replay Attack Demonstration The system demonstrates reuse of captured signals
- Use Case 5: Relay Attack Demonstration The system demonstrates real-time signal forwarding

- **Use Case 6: Secure Authentication** The system implements and validates a rolling code-based authentication mechanism to prevent unauthorized access

These functional requirements establish a clear understanding of system behavior and define the operational scope for implementation and evaluation.

3.2 Design constraints

The proposed system must operate within a set of clearly defined constraints that guide the design process and ensure that the final solution is practical, secure, and feasible. These constraints were identified during the initial planning phase and are aligned with both technical requirements and socio-economic considerations. Each constraint is measurable and directly influences design decisions, trade-offs, and evaluation criteria.

Table 3. Design Constraints

Constraint Type	Constraint Description	Justification and Relevance
Technical – Communication Reliability	The system must achieve $\geq 95\%$ successful transmission and reception rate within the operating range.	RF communication is inherently unreliable due to noise and interference. Ensuring high reliability is essential for correct system operation and user trust.
Technical – Response Time	The system must respond to user commands within ≤ 200 ms under normal conditions.	Fast response is required to maintain usability and user experience in real-time applications such as vehicle access.

Technical – Security (Replay Resistance)	The system must prevent reuse of previously transmitted signals.	Replay attacks are a major vulnerability in static RF systems. This constraint ensures that the implemented rolling code mechanism provides effective protection.
Technical – Synchronization Robustness	The system must tolerate limited missed transmissions using a bounded synchronization window.	Wireless environments may cause packet loss. A synchronization mechanism is required to maintain system functionality without compromising security.
Economic – Cost Efficiency	The total system cost must remain within a low-cost range (approximately QAR 110–180).	The solution must be affordable for practical deployment and competitive with existing market alternatives.
Usability – System Simplicity	The system must provide clear and simple user interaction through basic inputs and outputs (buttons, LEDs, buzzer).	Maintaining simplicity ensures ease of use and accessibility for non-technical users.
Environmental – Energy Efficiency	The system must operate using low-power components suitable for embedded applications.	Energy efficiency is important for battery-operated devices and supports sustainable system design.

These constraints define the boundaries within which the system is designed and implemented. They reflect the need to balance security, performance, cost, and usability. For example, improving security often increases computational

complexity, which may impact response time and power consumption. Similarly, enhancing reliability may require additional mechanisms such as synchronization windows, which introduce trade-offs between usability and strict security enforcement.

The selected constraints are realistic, measurable, and directly applicable to the system. They also serve as a basis for testing and evaluation in later sections, where the system performance is validated against these requirements.

Justification of Constraints

- **Communication Reliability and Response Time:** Reliable communication is essential for both system functionality and accurate evaluation. Without consistent signal transmission, it would be difficult to distinguish between system failure and attack effects.
- **Security Constraint:** Security is the core objective of the project. The system must clearly demonstrate both vulnerability and protection, making this constraint critical for validating the effectiveness of the proposed solution.
- **Cost Constraint:** Keeping the system affordable ensures that it can be widely adopted for educational purposes and potential real-world applications, especially as a retrofit solution.
- **Power Efficiency Constraint:** Since the system is designed for embedded and potentially portable environments, efficient power usage is necessary to maintain reliability and sustainability.
- **Ethical and Safety Constraint:** Because the project involves simulating attacks, it is essential to ensure that all activities are conducted responsibly and do not impact real-world systems or violate regulations.

These constraints define the operational boundaries of the system and ensure that the final design remains balanced in terms of performance, security, cost, and ethical responsibility

3.3 Design Standards

The proposed system is developed in alignment with relevant engineering and industry standards to ensure safety, reliability, interoperability, and compliance with accepted practices. Although the project is implemented as an academic prototype, adhering to these standards ensures that the design reflects real-world engineering requirements and can be extended toward practical applications.

Table 4. Design Standards Used in the Project

Standard Category	Standard / Guideline	Application in the Project
Wireless Communication Standards	ISM Band Regulations (433 MHz RF)	Ensures that RF modules operate within allowed frequency bands and comply with regional communication regulations.
Embedded System Design Standards	General Embedded System Design Practices	Guides efficient use of microcontrollers, memory, and real-time processing for reliable system operation.
Security Design Principles	Basic Cryptographic Principles (Confidentiality, Integrity, Freshness)	Applied through rolling code mechanism using counter, transformation function, and shared secret to prevent replay attacks.
Software Development Standards	Modular Programming and Code Structuring	Ensures readability, maintainability, and scalability of embedded software components.
Safety and Low-Voltage Design	Low-Voltage Electrical Safety Practices	Ensures safe operation of circuits and components used in

		prototyping (e.g., ESP32, RF modules, LEDs, buzzer).
Testing and Validation Standards	Functional and Performance Testing Methodologies	Used to verify system reliability, response time, and security effectiveness under different scenarios.
Ethical Engineering Standards	Controlled Security Testing Guidelines	Ensures that replay and relay attacks are demonstrated ethically within a controlled and safe environment.

Justification of Standards

The selection of design standards in this project ensures that the system is not only functional but also aligned with widely accepted engineering practices, enhancing its reliability, safety, and potential for real-world application.

- **Wireless Communication Standards (ISM Band, FCC Part 15):** These standards ensure that the system operates within regulated frequency bands, such as the 433 MHz ISM band, without causing harmful interference to other devices. Compliance with these guidelines is essential for both legal operation and ethical use of wireless communication technologies.
- **Security Standards (ISO/IEC 27001 – Principles):** Although the system is a prototype, it incorporates key security principles such as data integrity, authentication, and protection against replay attacks. Applying these principles aligns the design with modern cybersecurity practices and ensures that security is considered as a fundamental aspect of system development.
- **Embedded System Standards (IEEE Guidelines):** General embedded system design guidelines support proper system architecture, modularity, testing, and debugging. These practices contribute to a well-structured and reliable

implementation, ensuring that the system operates efficiently under real-time constraints.

- **Software Development Best Practices:** The use of modular programming, structured logic, and clear data handling improves code maintainability, scalability, and debugging efficiency. This is essential for managing system complexity and supporting future enhancements.
- **Testing and Validation Practices:** Standard testing methodologies are applied to verify system functionality, performance, and security. This ensures that the system meets its defined requirements and constraints under different operating conditions.

By adhering to these standards and best practices, the project ensures that the developed solution is safe, reliable, and aligned with professional engineering principles, thereby increasing its credibility and applicability in real-world scenarios.

3.4 Professional Code of Ethics

The development of a secure wireless vehicle access system involves several ethical, legal, and societal considerations, particularly because the project includes the simulation of security attacks such as replay and relay. These aspects require careful handling to ensure that the work is conducted responsibly, safely, and in alignment with professional engineering standards.

Ethical, Legal, and Security Considerations

A key ethical concern in this project is the implementation of attack techniques. Replay and relay attacks are included strictly for educational and research purposes and are performed only on controlled prototype systems developed by the project team. No external or real-world systems are targeted. This approach aligns with responsible cybersecurity practices, where vulnerabilities are studied to improve system security rather than exploit it.

From a legal perspective, wireless communication must comply with regulations governing RF transmission. All experiments are conducted within controlled

environments using standard frequency bands to avoid interference with other devices and ensure compliance with applicable communication guidelines.

From a societal perspective, the project contributes to enhancing public safety and awareness by highlighting vulnerabilities and demonstrating practical security solutions. This helps build trust in wireless systems and promotes responsible technology usage

Stakeholder Considerations

The project involves multiple stakeholders with different interests:

- Users: Require secure, reliable, and easy-to-use systems
- Engineers/Developers: Must design systems that prioritize safety, security, and efficiency
- Society: Benefits from improved security, reduced risk of unauthorized access, and increased cybersecurity awareness

Balancing these interests requires designing a system that is secure while remaining practical, cost-effective, and user-friendly

Diversity, Equity, Inclusion, and Accessibility (DEIA)

Although the system itself is not directly designed for accessibility-specific applications, DEIA principles are reflected in the team collaboration and development process:

- Inclusion: All team members actively contributed to discussions, design decisions, and implementation tasks, ensuring that every perspective was valued.
- Diversity: The team consists of members from different academic and personal backgrounds, enabling a variety of ideas and problem-solving approaches.
- Equity: Tasks and responsibilities were distributed fairly, ensuring equal opportunities for learning and contribution.

- **Accessibility:** Documentation and system design were developed in a clear and understandable manner, making the project accessible to a wider academic audience.

These practices foster a respectful and collaborative environment, aligning with institutional and professional expectations.

Application of ACM/IEEE Code of Ethics

The project follows selected principles from the ACM and IEEE Code of Ethics, applied specifically to its context:

- **Public Interest (ACM 1.1 / IEEE Principle 1):** The project prioritizes user safety by addressing real-world security vulnerabilities and proposing practical solutions
- **Avoid Harm (ACM 1.2):** Attack demonstrations are conducted in controlled environments to prevent misuse or unintended consequences
- **Honesty and Transparency (ACM 1.3 / IEEE Principle 3):** System performance, limitations, and vulnerabilities are reported accurately and objectively
- **Quality of Work (IEEE Principle 3):** The system is carefully designed, tested, and documented to ensure reliability and correctness
- **Respect for Privacy and Security (ACM 1.6):** The project emphasizes secure communication and protection against unauthorized access
- **Professional Responsibility (IEEE Principle 6):** The team adheres to ethical research practices, university guidelines, and safety procedures throughout the project

Ethical Decision-Making and Responsibility

Ethical considerations were addressed throughout the project by:

- Conducting all tests in controlled and authorized environments
- Clearly documenting the purpose and limitations of attack demonstrations
- Focusing on improving system security rather than exploiting vulnerabilities

Table 5: Summary of Ethical Issues and Responsibilities

Ethical Issue / Concern	Description	Responsibility / Action Taken
Attack Simulation (Replay & Relay)	Implementation of attack techniques could be misused if not controlled	Conducted only on project-owned prototype systems in controlled environments for educational purposes
Unauthorized Access Risk	Demonstrated system vulnerabilities may highlight real-world weaknesses	Focused on improving security and clearly documenting limitations and countermeasures
Wireless Communication Compliance	RF signals may interfere with other devices if not properly managed	Operated within standard ISM frequency bands and performed testing in controlled settings
Data Integrity and Security	Risk of accepting unauthorized or manipulated signals	Implemented rolling code-based authentication to ensure message validity and prevent replay attacks
User Safety and Trust	Insecure systems may lead to theft or misuse	Designed the system to enhance security and increase user confidence in wireless technologies
Honesty in Reporting	Risk of overstating system performance or hiding limitations	Reported all results, limitations, and trade-offs transparently and accurately
Team Collaboration (DEIA)	Potential imbalance in participation or decision-making	Ensured inclusive collaboration, fair task

		distribution, and respect for diverse perspectives
Professional Conduct	Need to follow ethical engineering and research practices	Adhered to ACM/IEEE guidelines, university policies, and safe laboratory practices

By integrating ethical considerations into both the design and implementation phases, the project ensures responsible engineering practice while addressing a critical real-world security problem.

3.5 Assumptions

The following assumptions are made during the planning and design phases of the project. These assumptions represent external conditions outside the direct control of the system design, which are expected to hold during implementation and testing. Any deviation from these assumptions may affect system performance and evaluation results.

- It is assumed that wireless communication operates within a stable indoor environment, where signal attenuation remains within the expected short-range limits.
- It is assumed that external RF interference from nearby devices (e.g., other RF transmitters) is limited during testing and does not significantly disrupt communication.
- It is assumed that hardware components function according to manufacturer specifications, including RF modules and microcontrollers.
- It is assumed that a stable and sufficient power supply is available during testing and demonstration to support continuous system operation.
- It is assumed that testing is conducted in a controlled environment, where replay and relay attack demonstrations can be safely performed without affecting external systems.
- It is assumed that project resources, including time, laboratory tools, and equipment, are available as planned throughout the development process.

- It is assumed that users interact with the system under normal operating conditions, without introducing unexpected misuse scenarios beyond the defined project scope.

These assumptions are monitored throughout the implementation and testing phases. If any assumption proves invalid, appropriate adjustments are made to ensure that the system continues to meet its design objectives under realistic conditions.

4. Project Plan

4.1 Project milestones

Table 6 Milestones of the project

Milestone	Description	Deliverables
Project Planning	Define project goals, identify hardware components, and software requirements. Create the system architecture.	Finalized project plan, system diagram, and requirements document.
Key-Fob Transmitter Prototype	Build and test the key fob (ESP32) to send lock/unlock signals. Test button functions and power supply.	Working transmitter circuit sending RF signals.
Car Receiver Prototype	Build the receiver (ESP32 + servo motor) to control the lock. Verify lock/unlock movement and signal reception.	Receiver circuit controlling servo for lock mechanism.
Attack Demonstration	Implement replay and relay attacks using two ESP32 boards to demonstrate system vulnerability.	Working replay and relay attack test setup and results.
Security Protocols Implementation	Add rolling code and challenge-response authentication using Encryption/Decryption protocol.	Secure RKE system prototype with verified communication.
Testing and Evaluation	Test secure system performance and confirm attacks no longer	Test report, performance evaluation, and security analysis.

	work. Compare insecure vs. secure results.	
Final Documentation & Presentation	Compile final report, prepare slides, and demo project to supervisor.	Final report and presentation.

4.2 Project timeline

Commented [QmM1]: Fill the missing parts of the table below.

Table 1 Project timeline

Task Number	Task Description	Assigned To	Start Date	End Date	Duration	Deliverables/ Outcome
1	Project Planning, define project goals, identify hardware components, and software requirements. Create the system architecture.	To all members	2/10/2025	16/10/2025	2 weeks.	Project plan and requirements document.
2	System Design, prepare system architecture diagram and communication flow between transmitter and receiver.	Muneera & Latifa	10/10/2025	16/10/2025	6 days.	System design and workflow diagram.
3	Prototype Development Key fob (Transmitter), design and test basic transmitting functionality.	Aldana & Amna	17/11/2025	17/11/2025	2 hours.	Working on key fob prototype.
4	Prototype Development Car Unit (Receiver), design and test basic receiving functionality and output control.	Aldana & Amna	17/11/2025	17/11/2025	2.5 hours.	Working car unit prototype.

5	Attack Demonstration, implement and test replay and relay attack scenarios to show system vulnerability.	Muneer a & Latifa	3/3/2026	10/3/2026	1 week.	Attack test setup and results.
6	Security Mechanism Design, develop rolling code and authentication protocols logic.	To all members.	15/3/2026	25/3/2026	10 days.	Security algorithm design.
7	Secure System Implementation, integrate security logic into the transmitter and receiver for secure communication.	Muneer a & Amna	1/4/2026	16/4/2026	15 days.	Secure prototype with verified communication.
8	Testing and Evaluation, compare performance between insecure and secure versions, record results, and analyze data.	Latifa & Aldana	17/4/2026	22/4/2026	5 days.	Performance report and data analysis.
9	Documentation and Review, write temporary and final report sections based on progress and feedback from supervisor.	To all members.	22/4/2026	29/4/2026	1 week.	Draft report ready for review.
10	Final Presentation Preparation, Prepare slides and demo video for	To all members.	25/4/2026	30/4/2026	5 days.	Final presentation and project demo.

	final evaluation and presentation.					
--	------------------------------------	--	--	--	--	--

4.3 Anticipated risks

The potential risks and the ways to minimize the effect is summarized in the table below.

Table 2. Potential Risks

Risk event	Approach to minimizing the effect on project success
Hardware component failure	Keep spare components; test each module individually before integration to detect issues early.
Cryptography implementation errors	Conduct thorough debugging, perform multiple security tests, review code and algorithms to ensure accuracy.
Safety hazard	Follow appropriate laboratory safety procedures and verify circuit connections before powering the system.
Testing environment limitations	Perform multiple trials under different conditions; identify and document anomalies for correction.
Team coordination issues	Hold weekly meetings to track progress; assign tasks and responsibilities to maintain consistent communication.
Data or file loss	Use cloud storage or GitHub for regular backups and version control to avoid losing important project files.
Time management issues or project delays	Follow the project timeline closely, review progress weekly, and prioritize critical tasks before minor improvements.

5. Solution design

This section presents the detailed design of the proposed secure wireless vehicle access system. The design is developed to satisfy the functional requirements while operating within defined constraints related to performance, cost, reliability, and security.

The solution follows an iterative engineering design approach, where multiple alternatives were evaluated and refined based on practical testing and real-world limitations of RF communication. In particular, the security mechanism was selected after evaluating more complex approaches and identifying the need for a lightweight and reliable solution suitable for one-way communication.

The system is structured into three progressive stages that demonstrate the full security lifecycle of wireless access systems.

5.1 Design Overview

The system adopts a three-stage architecture to clearly illustrate the transition from vulnerable to secure wireless communication:

- **Stage 1 (Baseline System):** A basic wireless communication system using static RF signal transmission to perform lock and unlock operations. This stage intentionally does not include security mechanisms in order to replicate real-world vulnerabilities found in simple RKE systems.
- **Stage 2 (Attack Models):** Replay and relay attack modules are introduced to exploit weaknesses in the baseline system. These modules demonstrate how intercepted signals can be reused or forwarded to gain unauthorized access, highlighting the limitations of static communication.
- **Stage 3 (Secure System – Rolling Code):** A rolling code-based authentication mechanism is implemented to enhance security. This approach uses a counter, a transformation function, and a shared secret key to ensure that each transmitted signal is unique and cannot be reused. The receiver independently validates incoming messages and enforces replay protection and synchronization.

5.2 High-Level Architecture

The proposed solution is organized as a modular wireless vehicle access system composed of interacting components that operate across three functional layers: the legitimate access system, the attack modules, and the secure authentication mechanism. This architecture supports the project's three-stage approach, where system behavior evolves from baseline operation to attack demonstration and finally to a secure implementation.

The legitimate access system consists of two primary components:

- The key unit, which captures user input (lock/unlock commands) and transmits RF signals
- The vehicle unit, which receives signals, processes commands, and controls system outputs (LEDs and buzzer)

The attack layer operates by interacting directly with the communication channel between the key and the vehicle. Instead of being a separate stage after system operation, the attack modules are integrated into the signal path to demonstrate real-world vulnerabilities:

- The **replay module** captures and stores transmitted signals, then retransmits them later
- The **relay module** forwards signals in real time between two devices to extend communication range

The secure system layer enhances the communication process by replacing static transmission with a rolling code-based authentication mechanism. In this layer:

- The key unit generates a unique encoded message using a counter, a transformation function, and a shared secret key
- The vehicle unit independently computes the expected value and verifies the received message
- Replay protection and synchronization mechanisms ensure that only valid and fresh signals are accepted

The interaction between these components occurs through the wireless communication channel, which serves as the central interface of the system. The architecture highlights

how vulnerabilities arise within this channel and how the secure mechanism mitigates them.

This high-level organization provides a clear abstraction of system components, their roles, and their interactions, forming the foundation for detailed hardware and software design in subsequent sections

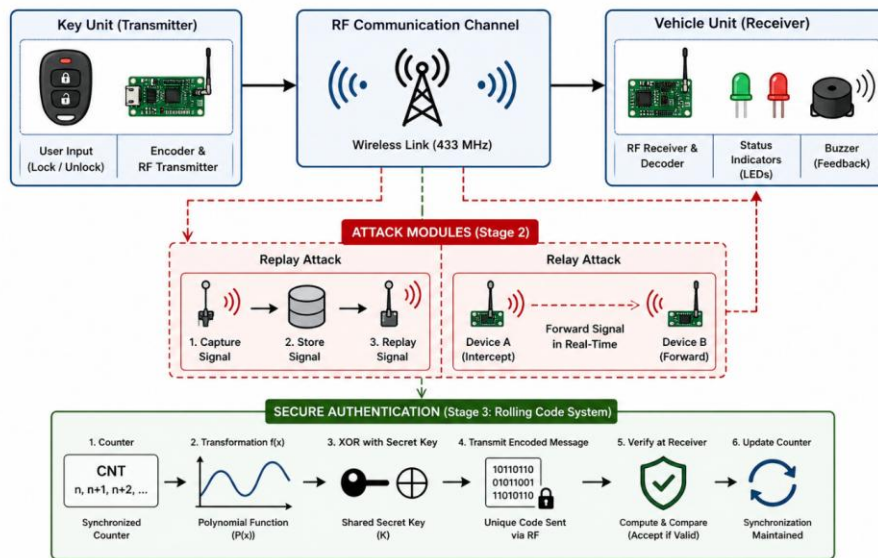


Figure 2. High-Level Architecture of the Proposed System

Key Components and Roles

Remote Key Unit: The remote key acts as the user-side device responsible for initiating lock and unlock operations. In the baseline stage, it transmits fixed RF signals corresponding to user commands. In the secure stage, it generates dynamic transmission codes using a rolling code mechanism based on a synchronized counter

and a shared secret key, ensuring that each transmitted signal is unique and resistant to replay.

Vehicle Unit: The vehicle unit receives wireless signals and determines whether access should be granted or denied. In Stage 1, it performs a direct comparison between the received signal and stored static codes. In the secure stage, it independently computes the expected rolling code using its internal counter and shared key, and validates the received signal within an acceptable synchronization window before activating any output.

Attack Demonstration Modules: These modules are used exclusively in controlled testing environments. The replay attack device captures and stores valid RF signals and retransmits them to demonstrate unauthorized access. The relay attack system forwards signals in real time between the key unit and the vehicle unit, extending the communication range and simulating real-world attack scenarios.

Output Feedback Unit: The output feedback unit consists of LEDs and a buzzer that provide clear visual and auditory confirmation of system actions. This approach replaces mechanical components such as servo motors, improving system reliability, reducing power consumption, and maintaining focus on wireless communication behavior and security analysis.

Normal Operation (Stage 1):

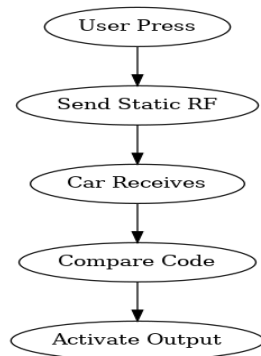


Figure 3: Stage 1 Static Communication Workflow

Figure 3 presents the workflow of the baseline system, where a user initiates a command that is transmitted as a static RF signal to the vehicle unit. The vehicle processes the received signal and activates the output if the code matches predefined values. This simple design demonstrates the lack of authentication, making the system vulnerable to signal-based attacks

Attack Scenarios (Stage 2):

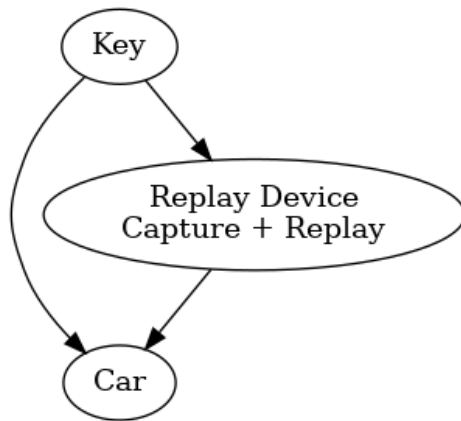


Figure 4: Replay Attack Scenario

Figure 4 demonstrates how a replay attack is performed. The attacker captures a valid RF transmission from the key and stores it. At a later time, the attacker retransmits the same signal to the vehicle, which accepts it as a legitimate command due to the absence of dynamic authentication mechanisms.

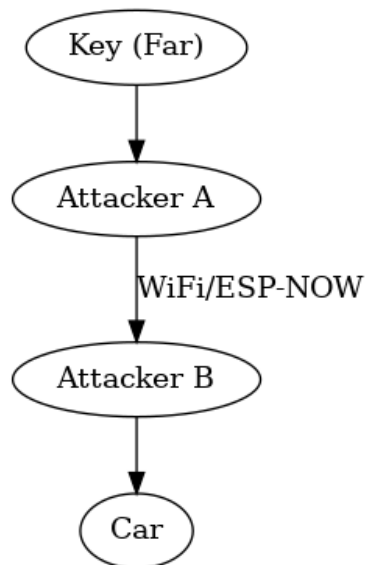


Figure 5: Relay Attack System Using Dual Devices

Figure 5 illustrates a relay attack, where two attacker devices cooperate to extend the communication range between the key and the vehicle. One device captures the signal near the key and forwards it in real time to another device located near the vehicle. This allows unauthorized access even when the legitimate key is physically distant.

Rolling Code Authentication (Stage 3)

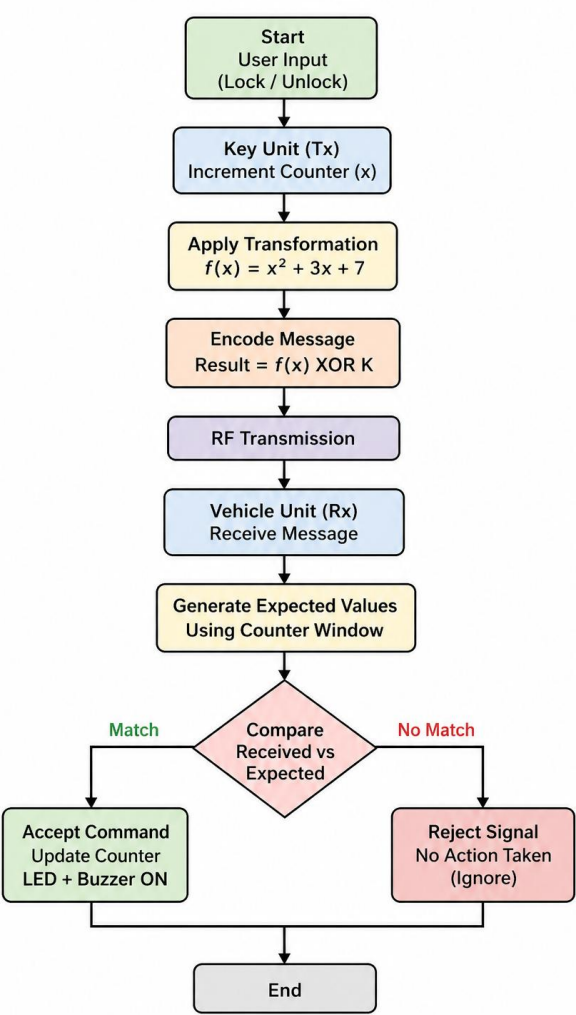
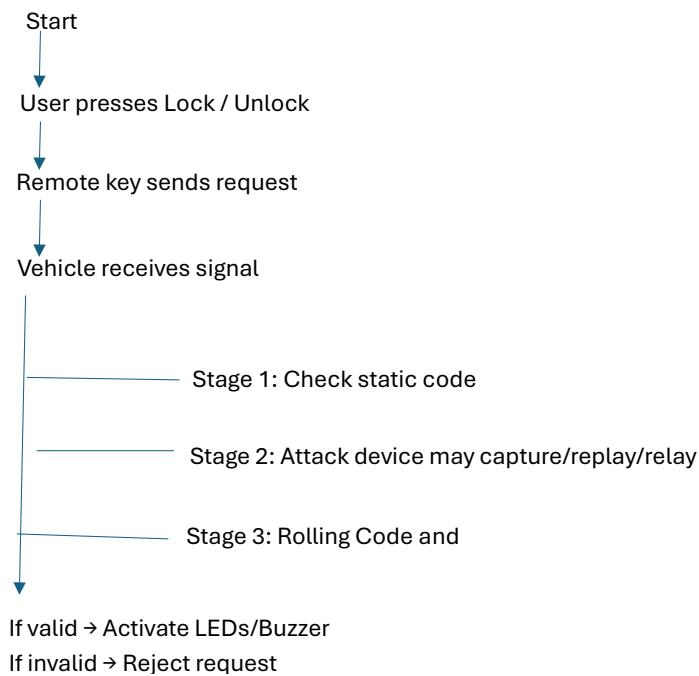


Figure 6 shows the system uses a rolling code mechanism to ensure secure communication between the key and the vehicle. When the user initiates a command, the key increments an internal counter and generates a dynamic

authentication value using a transformation function and a shared secret key. The encoded message is then transmitted to the vehicle.

Upon receiving the message, the vehicle independently computes the expected authentication value using its own counter and a predefined synchronization window. The received value is compared with the expected values, and access is granted only if a valid match is found. This process ensures that each transmission is unique and prevents reuse of previously captured signals, providing effective protection against replay attacks.

High-Level Software Logic



This architecture separates the system into clear functional blocks, making the design easier to implement, test, and evaluate. It also allows direct comparison between vulnerable and secure operation modes.

5.3 Design Alternatives

The design process followed an iterative engineering approach, where multiple potential solutions were evaluated against key project constraints, including security, cost, reliability, development complexity, and suitability for embedded implementation. Since the project aims to demonstrate both system vulnerabilities and practical countermeasures, the selected design must support experimental analysis as well as real-world feasibility.

Alternative 1: NFC-Based Secure Access System

This design uses short-range wireless communication, such as Near Field Communication (NFC), to perform vehicle access control. Due to its extremely limited range, NFC inherently reduces the risk of long-range replay and relay attacks and supports secure authentication mechanisms.

However, this approach is not suitable for this project because it does not accurately represent real-world Remote Keyless Entry (RKE) systems, which rely on RF communication. The short communication range limits its ability to demonstrate realistic attack scenarios. Additionally, it requires specialized hardware, increasing system cost and reducing flexibility.

Alternative 2: Proprietary Encrypted RF System

This design implements secure RF communication using cryptographic algorithms from the outset. It provides strong protection against common attacks and reflects modern commercial keyless entry systems.

However, this approach was not selected because it prevents clear demonstration of system vulnerabilities. Starting directly with a secure system eliminates the ability to illustrate how replay and relay attacks exploit weak designs. It also increases system

complexity and computational overhead, making it less suitable for low-cost embedded platforms and educational purposes.

Alternative 3: Progressive RF System with Rolling Code Security

The selected design uses a basic RF communication system initially and then progressively introduces attack demonstrations and security enhancements. This approach aligns with the three-stage structure of the project: baseline operation, attack demonstration, and secure system implementation.

In the secure stage, a rolling code-based authentication mechanism is used, where each transmitted signal is dynamically generated using a counter, a transformation function, and a shared secret key. This ensures that transmitted messages are unique and cannot be reused.

This design was selected because it:

- Enables clear demonstration of vulnerabilities (replay and relay attacks)
- Provides a practical and lightweight security mechanism suitable for embedded systems
- Maintains low cost and ease of implementation
- Supports real-time operation under RF communication constraints
- Aligns with the project objectives and evaluation criteria

Table 9. Comparison of Design Alternatives

Criteria	NFC System	Encrypted RF System	Progressive RF with Rolling Code
Security Level	High	Very High	Moderate–High
Replay Attack Resistance	High	High	High
Relay Attack Resistance	High (range-limited)	Moderate	Limited
Cost	High	High	Low
Ease of Implementation	Moderate	Complex	Simple

Suitability for Demonstration	Low	Low	High
Realism (RKE Systems)	Low	High	High

The selected design provides the best balance between cost, security analysis, implementation feasibility, and educational value. It also enables a clear comparison between insecure and secure wireless access systems, which is central to the purpose of this project.

5.4 Hardware/Software Used

This section presents the specific hardware and software components used in the design and implementation of the proposed system. The components were carefully selected based on cost, availability, ease of integration, and suitability for real-time embedded applications. The design focuses on low-cost hardware while ensuring sufficient capability to implement wireless communication and rolling code authentication.

Table 10. Needed HW/SW

HW/SW Details	Platform	Justification
ESP32 Microcontroller	Embedded Processing Platform	Acts as the main processing unit for both key and vehicle units. Selected for its low cost, sufficient processing capability, built-in peripherals, and ease of development using Arduino IDE.
433 MHz RF Transmitter (FS1000A)	Wireless Communication Platform	Used to transmit wireless signals from the key unit. Chosen for its low cost, availability, and suitability for

		demonstrating RF-based vulnerabilities.
433 MHz RF Receiver (XY-MK-5V)	Wireless Communication Platform	Used in the vehicle unit to receive RF signals. Compatible with the transmitter and provides reliable short-range communication.
Push Buttons	Human–Machine Interface	Used to generate user input commands (lock/unlock). Simple, reliable, and easy to integrate with the microcontroller.
LEDs	Output / Indication Platform	Provide visual feedback indicating system states (lock/unlock, success/failure). Low power and easy to interface.
Buzzer	Output / Indication Platform	Provides audio feedback to enhance system interaction and confirm operations.
Breadboard & Jumper Wires	Prototyping Platform	Used for hardware prototyping and circuit connections. Enables rapid development and testing.
Arduino IDE	Software Development Platform	Used for programming and debugging the system. User-friendly and widely supported for embedded development.
Embedded C/C++	Programming Language	Used to implement system logic, RF communication, and rolling code authentication. Efficient for real-time applications.

Serial Monitor	Debugging Tool	Used to monitor system behavior, verify communication, and assist in testing and troubleshooting.
-----------------------	----------------	---

Justification of Design Choices

- **ESP32 Microcontroller Selection:** The ESP32 microcontroller was selected due to its balance between processing capability, cost, and ease of development. It provides sufficient computational power to handle RF communication, timing-sensitive operations, and the rolling code authentication mechanism, while remaining suitable for real-time embedded applications.
- **433 MHz RF Communication Modules:** The RF transmitter (FS1000A) and receiver (XY-MK-5V) were chosen because they closely represent real-world low-cost RKE systems that are commonly vulnerable to replay and relay attacks. This makes them ideal for both demonstrating system weaknesses and evaluating security improvements. Their simplicity also allows direct control over signal behavior without relying on proprietary protocols.
- **Rolling Code Security Mechanism:** A rolling code approach was selected instead of a full cryptographic protocol to achieve a balance between security and practical reliability. This method introduces dynamic message generation using a counter and a shared key, preventing reuse of captured signals while maintaining low computational overhead suitable for embedded systems.
- **LEDs and Buzzer as Output Feedback:** LEDs and a buzzer were used instead of mechanical actuators (e.g., servo motors) to improve system reliability and reduce power consumption. This design choice simplifies the hardware and ensures that system evaluation focuses on communication performance and security behavior rather than mechanical operation.

- **Software Environment (Arduino IDE & Embedded C/C++):** The Arduino IDE and Embedded C/C++ were selected to enable efficient development, debugging, and implementation of real-time communication logic. This environment provides flexibility and strong library support while maintaining control over low-level operations required for RF signal handling and synchronization.
- **Use of Supporting Libraries and Tools:** Existing libraries for RF communication and system timing were used to reduce development complexity and improve reliability. This allows the design effort to focus on system architecture, attack modeling, and security implementation rather than low-level protocol development.

The combination of these hardware and software platforms provides a balanced solution that meets project requirements while supporting scalability, reliability, and effective demonstration of both vulnerabilities and secure communication mechanisms.

5.5 Hardware Design

This section presents the detailed hardware design of the proposed system. The design is structured into multiple subsystems that collectively enable wireless communication, attack demonstration, and secure authentication using a rolling code mechanism. The hardware design supports the three-stage architecture of the project: a baseline vulnerable system, attack models, and a secure system implementation.

5.5.1 Remote Key Unit

The remote key unit serves as the user interface of the system. It is responsible for generating lock and unlock commands based on user input and transmitting them wirelessly.

The design follows a simple input–process–transmit model. User inputs are captured through push buttons and processed by a microcontroller to generate corresponding command signals. These signals are then transmitted via a wireless communication module.

Two design approaches were considered: a simple one-way transmitter and an enhanced transmitter with rolling code capability. The one-way design is sufficient for demonstrating static-code vulnerabilities in Stage 1; however, it cannot provide secure authentication. Therefore, in Stage 3, the design evolves to incorporate a rolling code mechanism, where each transmitted signal is dynamically generated using a counter and a shared key. This ensures that every transmission is unique and prevents the reuse of captured signals.

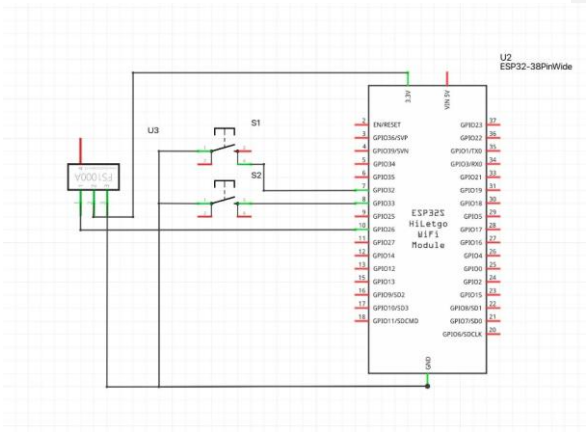
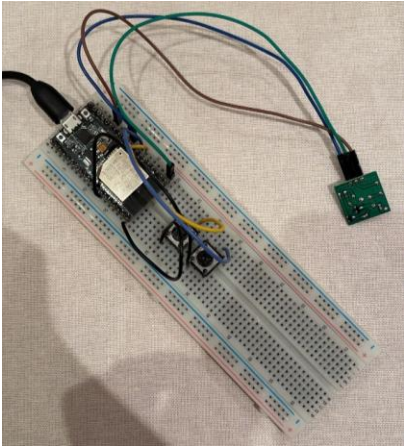


Figure 7: remote key unit

5.5.2 Vehicle Unit

The vehicle unit acts as the receiver and decision-making subsystem. It processes incoming wireless signals and determines whether to execute the requested action. The design includes a wireless receiver interface, a processing unit, and an output subsystem. In Stage 1, the system compares received signals with predefined static values. In Stage 3, it performs authentication using a rolling code mechanism by validating the received encoded signal against expected values generated from a synchronized counter and shared key.

The main design requirement is reliability and fast response. The system must correctly interpret signals, maintain synchronization with the transmitted codes, and provide consistent output under repeated operation.

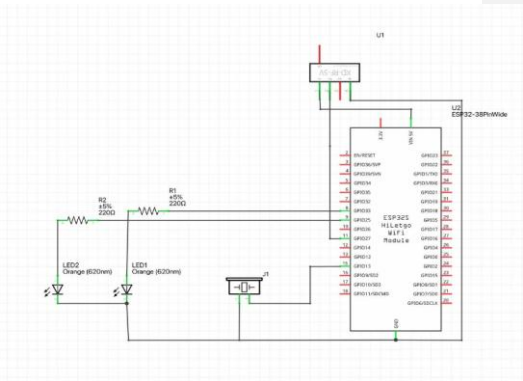
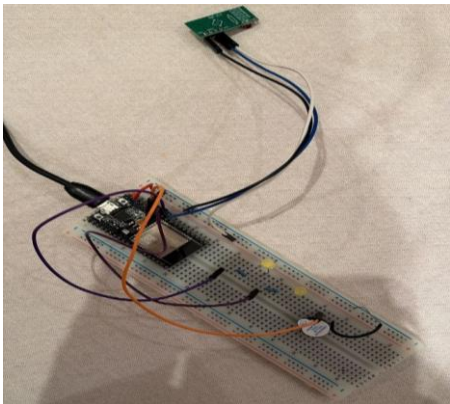


Figure 8: vehicle unit

5.5.3 Wireless Communication Subsystem

The wireless communication subsystem enables data exchange between the remote key and the vehicle unit. Several communication technologies were evaluated, including short-range communication and encrypted protocols.

Short-range technologies provide strong inherent security but do not represent real-world keyless entry systems. Encrypted protocols offer better protection but introduce complexity and reduce transparency for demonstration purposes.

The selected approach uses sub-GHz RF communication, which closely reflects real-world vulnerable systems. This allows clear demonstration of replay and relay attacks before implementing secure mechanisms.

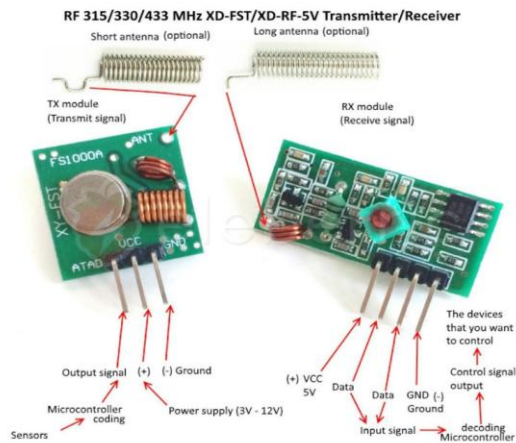


Figure 8: wireless communication subsystem

5.5.4 Replay Attack Hardware

The replay attack subsystem is designed to capture and reuse valid wireless signals. It consists of a receiver to capture signals, a processing unit to store them, and a transmitter to replay them.

This design demonstrates the weakness of static communication systems. Since no authentication is used, the system accepts repeated signals as valid commands.

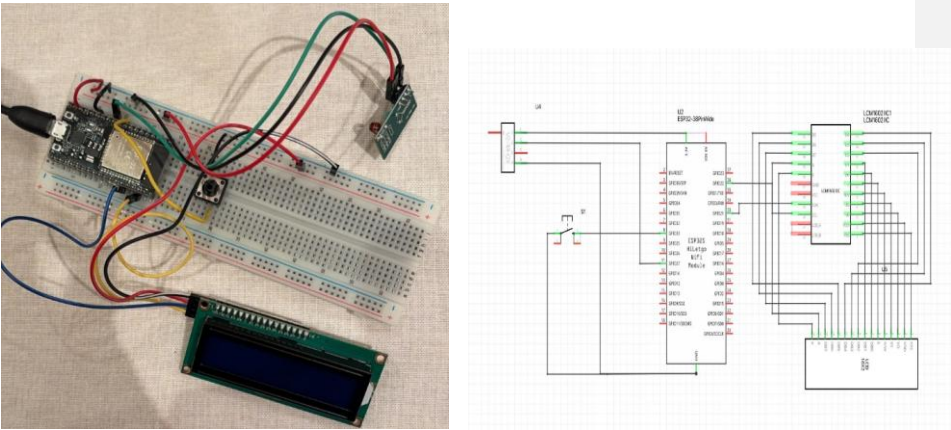


Figure 9: replay attack subsystem

5.5.6 Secure Bi-Directional Hardware Design

In Stage 3, the system transitions to a secure architecture that enhances communication using a rolling code mechanism. The key unit transmits dynamically generated authentication codes, while the vehicle unit receives and validates these signals.

Each transmission is based on an incremented counter combined with a transformation function and a shared secret key, ensuring that every message is unique and cannot be reused. The vehicle unit independently verifies the received signal and grants access only if a valid match is found within the synchronization window.

The trade-off in this design is a slight increase in processing and synchronization complexity compared to the baseline system. However, this is necessary to improve security and effectively prevent replay attacks while maintaining system reliability

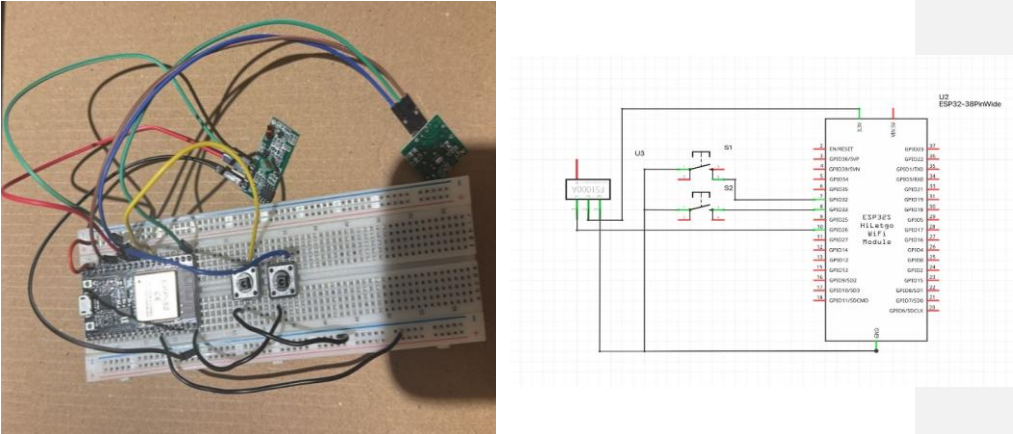


Figure 10: Secure Rolling Code Hardware Design

5.5.7 Output Feedback Subsystem

The output subsystem provides system feedback through visual and auditory indicators. Instead of using mechanical actuation, the design uses LEDs and a buzzer. This choice improves reliability, reduces power consumption, and simplifies the hardware design. It also ensures that the focus of the system remains on wireless communication and security rather than mechanical performance.

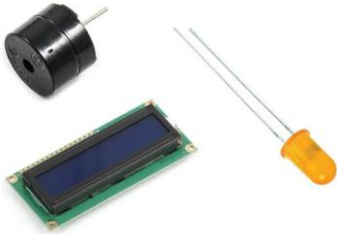


Figure 11: output subsystem

Summary of Hardware Design Choices

Table 11. Summary of Hardware Design Choices

Subsystem	Design Choice	Justification
Key Unit	ESP32 + RF Transmitter + Push Buttons	Provides a simple and reliable interface for generating user commands and transmitting RF signals. ESP32 offers sufficient processing capability for implementing rolling code logic.
Vehicle Unit	ESP32 + RF Receiver + Output Module (LED + Buzzer)	Enables reception, validation, and execution of commands. Supports rolling code verification while providing clear system feedback.
RF Communication	433 MHz RF Modules (FS1000A + XY-MK-5V)	Low-cost and widely available. Suitable for replicating real-world vulnerabilities and demonstrating RF-based attacks.
Replay Attack Module	RF Receiver + Storage + RF Transmitter	Allows capturing and retransmitting signals to demonstrate replay attacks under controlled conditions.
Relay Attack Module	Two RF Devices (Forwarding Mechanism)	Simulates real-time signal forwarding to extend communication range and demonstrate relay attacks.
Security Mechanism	Rolling Code (Counter + Transformation + Shared Key)	Provides dynamic authentication and prevents reuse of captured signals while maintaining low computational overhead.

Output Feedback	LEDs and Buzzer	Offers simple, reliable, and low-power indication of system states without mechanical complexity.
System Integration	Modular Hardware Design	Enables independent testing of subsystems and simplifies debugging and system expansion.

Overall, the hardware design provides a practical and modular platform for demonstrating the complete security lifecycle of wireless vehicle access systems: vulnerability, exploitation, and defense.

5.6 Software Design

This section presents the software design of the proposed system, focusing on both structural and behavioral aspects. The software is designed to control wireless communication, process signals, simulate attacks, and implement secure authentication using a rolling code mechanism. It is tightly integrated with the hardware components and follows a modular architecture to support the three operational stages of the project: baseline system, attack models, and secure system.

5.6.1 Software Architecture Overview

The software is organized into modular components, each responsible for a specific function:

- **Input Handling Module** – Reads user input (lock/unlock commands)
- **Communication Module** – Handles wireless transmission and reception
- **Processing and Control Module** – Executes decision-making logic and system flow
- **Attack Logic Module** – Implements replay and relay attack behavior for testing purposes

- **Authentication Module** – Implements rolling code generation and verification using a counter, transformation function, and shared key
- **Output Control Module** – Controls LEDs and buzzer for system feedback

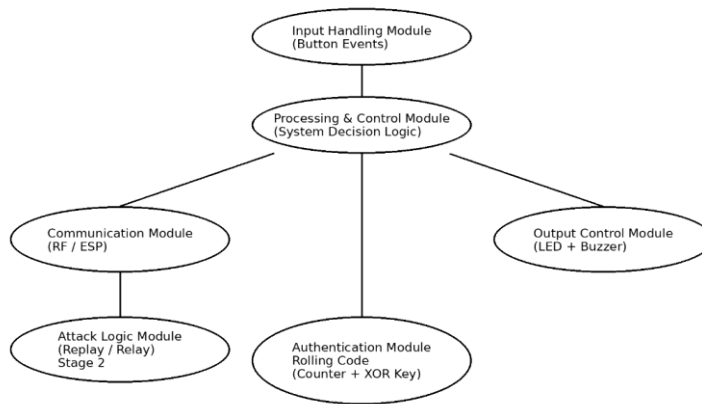


Figure 12: Software Architecture Block Diagram

5.6.2 Remote Key Software Design

The remote key software is responsible for detecting user input and transmitting commands.

Operation:

- Continuously monitors button states
- When a button is pressed:
 - Generates a command signal
 - Sends the signal via the communication module

In Secure Mode:

- Increments the internal counter for each operation
- Generates a unique rolling code using a shared secret key and transformation function
- Transmits the encoded signal to the vehicle
- Ensures that each transmission is different to prevent reuse of captured signals

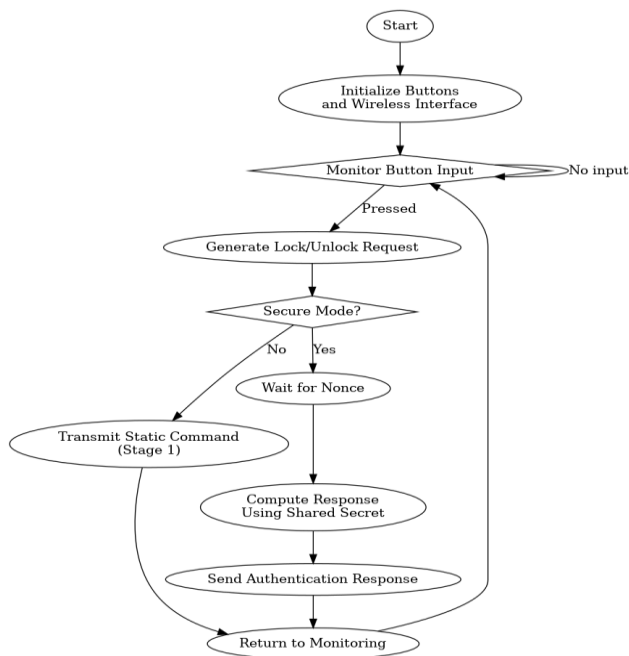


Figure 13: Remote Key Software Flowchart

5.6.3 Vehicle Unit Software Design

The vehicle unit software acts as the decision-making component.

Operation:

- Listens for incoming wireless signals
- Processes received data

In Stage 1:

- Compares received signal with stored codes
- Executes action if matched

In Stage 3:

- Receives the encoded rolling code from the key
- Generates expected codes based on the synchronized counter and shared key
- Compares the received code with the expected values within a synchronization window
- Verifies the correctness of the received code
- Grants or denies access based on the verification result

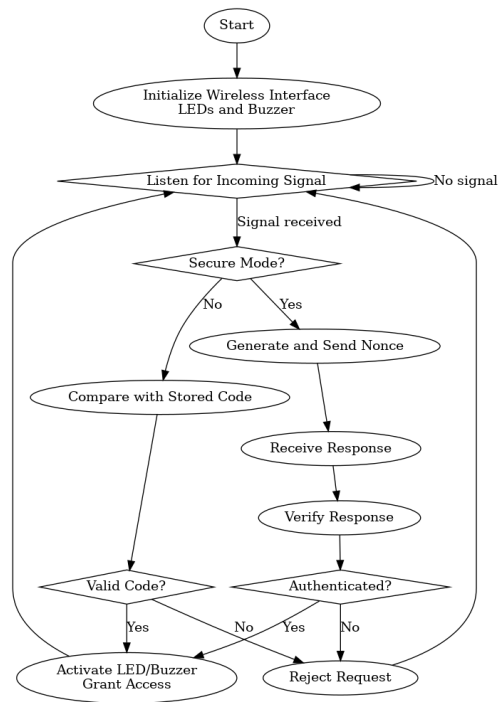


Figure 14: Vehicle Unit Software Flowchart

5.6.4 Relay Attack Software Logic

The relay attack software uses two devices communicating in real time.

Operation:

- Device A captures signal near key
- Sends data via wireless link (ESP-NOW)
- Device B receives and retransmits immediately

This creates a real-time signal extension.

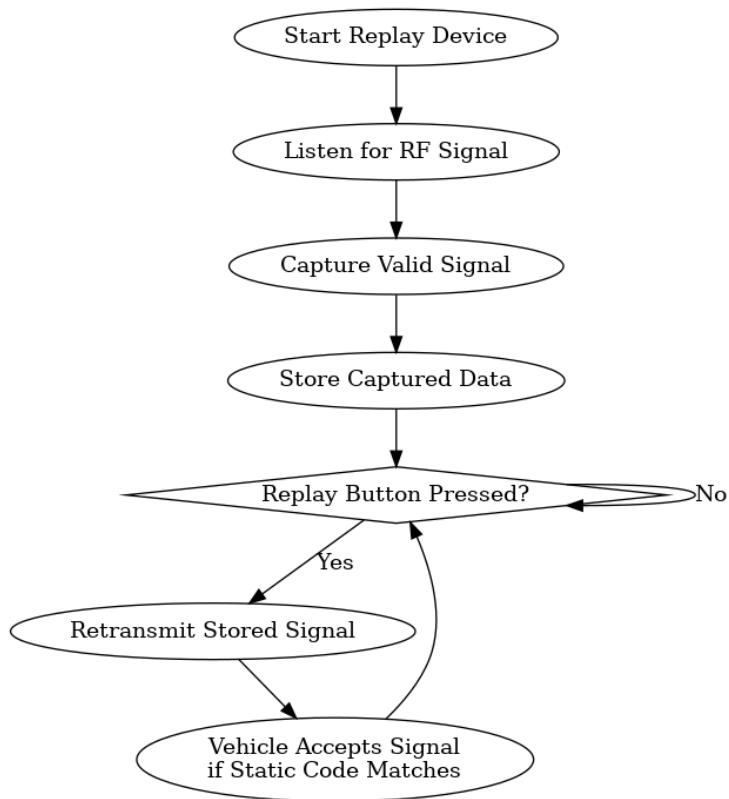


Figure 15: Relay Attack Software Logic

5.6.5 Secure Authentication Software

The secure system implements a rolling code authentication mechanism to ensure that each transmitted signal is unique and cannot be reused. This approach enhances security while maintaining low computational complexity suitable for embedded systems.

Operation:

1. Key generates a code using an internal counter
2. Counter is combined with a shared secret key using a transformation function (e.g., XOR)
3. Encoded value is transmitted to the vehicle
4. Vehicle computes the expected code using its own counter and shared key
5. If a match is found within an acceptable counter window, access is granted

Key Software Responsibilities:

- Maintain and increment the internal counter
- Generate encoded rolling code using transformation function
- Transmit encoded signal to the vehicle

Vehicle Software Responsibilities:

- Maintain expected counter value
- Generate expected codes for a defined counter window
- Compare received code with expected values
- Update counter upon successful validation
- Control system access (grant/deny)

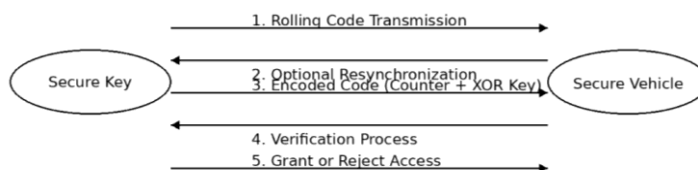


Figure 16: Rolling Code Authentication Sequence

5.6.6 System Behavior Summary

The software design supports three operational modes:

- **Stage 1:** Simple command transmission without security
- **Stage 2:** Signal interception and exploitation through replay and relay attacks
- **Stage 3:** Secure authentication using a rolling code mechanism and code verification

The modular design allows clear separation between these modes, making the system easier to test, analyze, and extend.

The software architecture provides a structured and modular approach to managing wireless communication, attack simulation, and secure authentication. It ensures reliable interaction with hardware components while maintaining flexibility for future improvements. The design effectively demonstrates both system vulnerabilities and enhanced security mechanisms within a unified framework.

6. Implementation

This section presents the actual implementation of the proposed system, including hardware realization, software deployment, wiring configurations, tools used, and practical challenges encountered. Unlike the design section, this part focuses on what was physically built and tested, aligned with the three-stage architecture: baseline system, attack models, and secure rolling code system.

6.1 System Implementation Overview

The system was implemented as a set of interconnected hardware modules:

- **Remote Key Unit**
- **Vehicle Unit**
- **Replay Attack Device**
- **Relay Attack System (Dual Devices)**
- **Secure Rolling Code System**

Each module was assembled on a breadboard using embedded components and connected through wireless communication.

6.2 Hardware Implementation and Wiring

Remote Key Unit (Actual Wiring)

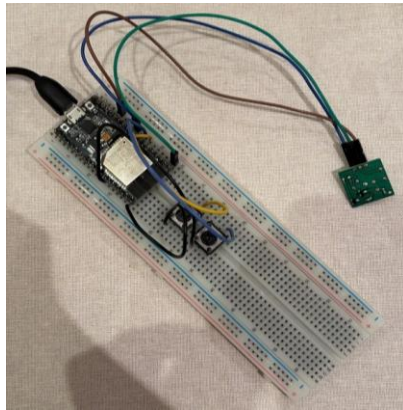


Figure 17: Remote Key Unit

Description:

- Push buttons connected to GPIO pins
- RF transmitter connected to digital output
- Power supplied via USB
- Microcontroller reads button state and sends signal

Vehicle Unit (Receiver + Output)

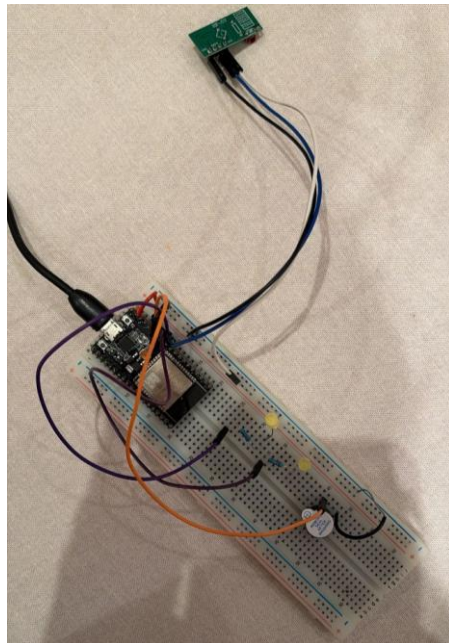


Figure 17: Vehicle Unit

Description:

- RF receiver connected to input pin
- LEDs connected via resistors
- Buzzer connected to output pin
- System activates output upon valid signal

Replay Attack Device

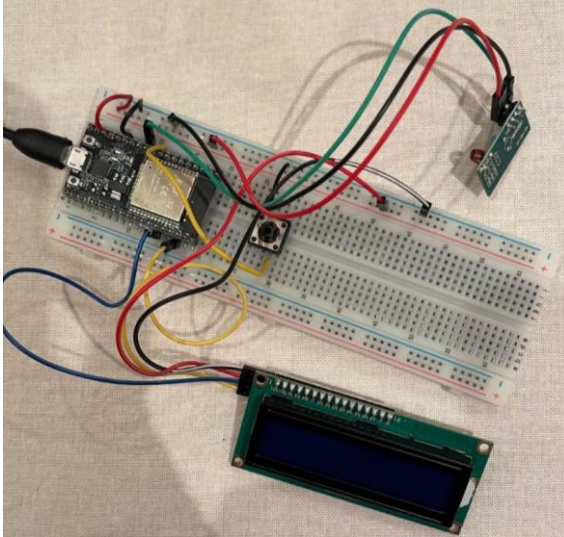


Figure 18: Replay Attack Device

Description:

- RF receiver captures signals
- Data stored in memory
- RF transmitter used to replay signal
- LCD displays captured status

Secure System (Rolling Code Authentication)

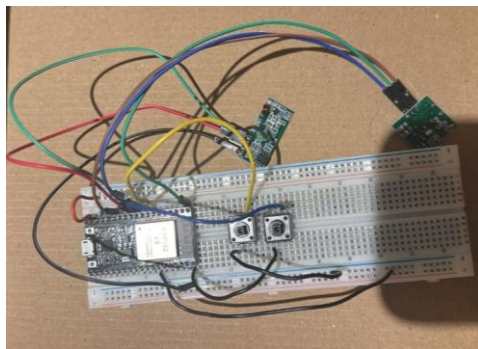


Figure 19: Secure System (Rolling Code Authentication)

Description:

- The key unit is equipped with a transmitter, while the vehicle unit includes a receiver for wireless communication
- The system uses a synchronized counter and a shared secret key to generate dynamic authentication codes
- Implements a rolling code authentication mechanism to ensure that each transmitted signal is unique and cannot be reused

6.3 Software Implementation

The system software was implemented using embedded programming techniques:

- The key module detects user input and generates wireless commands
 - The vehicle module receives signals, processes them, and controls system outputs
 - The attack modules simulate replay and relay attacks for testing purposes
 - The secure module implements rolling code authentication logic
- Communication protocols were implemented using:

- RF signal encoding and decoding
- ESP-NOW for relay communication
- Timing control for synchronization and counter management

Table 12.Tools and Frameworks Used

Tool / Platform	Purpose
Arduino IDE	Code development and uploading
Embedded C/C++	System programming
Serial Monitor	Debugging and testing
ESP-NOW Protocol	Low-latency device communication
RF Communication Libraries	Signal transmission and reception
Breadboard & Wiring Tools	Hardware prototyping

6.4 Novel Aspects of Implementation

The implementation introduces several unique features:

- Integration of both attack demonstration and security mechanisms within a single unified system
- Real-time demonstration of replay and relay attacks under controlled conditions
- Progressive transition from a vulnerable system to a secure system using a rolling code mechanism
- Low-cost embedded implementation of rolling code-based authentication suitable for resource-constrained devices

This approach provides both educational and practical value by enabling clear visualization of system vulnerabilities and their corresponding countermeasures within a single platform.

6.6 Implementation Challenges and Solutions

Challenge 1: RF Signal Interference

- **Problem:** Unstable signal reception due to environmental noise and interference
- **Solution:** Optimized antenna placement and minimized nearby noise sources to improve signal clarity

Challenge 2: Timing Synchronization

- **Problem:** Variations in signal timing affecting communication reliability and counter alignment

- **Solution:** Implemented timing control mechanisms and a synchronization window to maintain alignment between transmitted and expected rolling codes

Challenge 3: Signal Reliability

- **Problem:** Missed or inconsistent transmissions during RF communication
- **Solution:** Applied repeated transmission checks to improve delivery reliability

Challenge 4: System Complexity in Stage 3

- **Problem:** Managing rolling code generation, validation, and counter synchronization
- **Solution:** Modularized the authentication logic and separated encoding and verification processes to simplify implementation and debugging

The implementation successfully demonstrates a complete wireless security lifecycle, from vulnerable communication to secure rolling code authentication. The system validates the feasibility of low-cost embedded security solutions while highlighting real-world challenges and their practical resolutions.

7. Testing and Evaluation

This section presents a comprehensive evaluation of the implemented system through a series of structured tests. The objective is to verify that the prototype satisfies the defined functional requirements and design constraints, while also demonstrating the presence of vulnerabilities and the effectiveness of the implemented security mechanisms.

The system was tested across three main stages:

- **Stage 1:** Basic functionality (lock/unlock communication)
- **Stage 2:** Vulnerability demonstration (replay and relay attacks)
- **Stage 3:** Security validation using a rolling code authentication mechanism

7.1 Testing Overview

Testing was conducted using a test-case-driven approach, where each system functionality was validated through specific procedures and observed outputs. Both qualitative (behavioral) and quantitative (performance) results were collected.

The evaluation focused on:

- Correctness of system operation
- Reliability of wireless communication
- Vulnerability to replay and relay attacks
- Effectiveness of the rolling code authentication mechanism

7.2 Stage 1 – Functional Testing (Basic System)

Table 13. Test Cases

Test Case	Input Action	Expected Output	Actual Result
Unlock Operation	Press unlock button	LED blinks once + buzzer beeps once	Achieved
Lock Operation	Press lock button	LED blinks twice + buzzer beeps twice	Achieved
Signal Transmission	Send RF signal	Signal received and decoded	Achieved
Serial Monitoring	Trigger command	Display transmission logs	Achieved

Observations

- The system successfully transmitted and received RF signals using wireless communication.
- LED and buzzer outputs clearly represented system states:
 - Unlock → 1 blink + 1 beep
 - Lock → 2 blinks + 2 beeps

- Serial monitor output confirmed correct communication flow (e.g., “Unlock signal sent”, “Received unlock signal”, ID verification).
- The system response was fast and consistent, with no noticeable delay.

Evaluation

The system fully satisfied the functional requirements for basic operation. Communication reliability and output clarity were both achieved.

7.3 Stage 2 – Vulnerability Testin

A. Replay Attack Testing

Test Procedure

- Capture RF signal during legitimate key usage
- Store signal in attacker device memory
- Replay stored signal without original key
- Observe system response

Table 14. Result

Parameter	Expected Result	Actual Result
Signal Capture	Signal successfully captured	Achieved
Signal Storage	Signal stored in memory	Achieved
Replay Function	Stored signal reused	Achieved
System Response	System accepts replayed signal	Achieved

Observations

- Replay attack was successfully demonstrated.
- The system allowed previously captured signals to be reused without authentication.
- LCD interface provided clear feedback (e.g., “Capturing Signal”, “Stored in S1”, “Replaying S1”, “All Cleared”).

- System performance remained stable with minimal delay.

Evaluation

The results confirm a critical vulnerability in the baseline system. The absence of dynamic authentication mechanisms allows attackers to gain unauthorized access using replayed signals.

B. Relay Attack Testing

Test Procedure

- Place Device A near the legitimate key
- Place Device B near the vehicle
- Forward signal in real time using wireless link
- Observe vehicle response

Table 15. Result

Parameter	Expected Result	Actual Result
Signal Capture	Signal captured near key	Achieved
Real-Time Forwarding	Signal transmitted between devices	Achieved
System Response	Vehicle accepts forwarded signal	Achieved

Observations

- Relay attack successfully extended the communication range between the key and vehicle.
- The vehicle responded as if the key was physically nearby.
- No authentication mechanism prevented the forwarded signal.

Evaluation

The system is vulnerable to relay attacks, demonstrating that proximity-based trust alone is insufficient for security.

7.4 Stage 3 – Security Testing

Test Procedure

- Initiate unlock request from the key
- Key increments the internal counter
- Key generates an encoded rolling code using a transformation function and shared key
- Encoded signal is transmitted to the vehicle
- Vehicle computes expected codes within a synchronization window
- Vehicle compares received code with expected values
- System grants or denies access based on verification result

Table 16. Result

Parameter	Expected Result	Actual Result
Counter Increment	Counter increases correctly with each transmission	Achieved
Code Generation	Unique encoded rolling code generated for each command	Achieved
Code Verification	Only valid rolling codes within synchronization window are accepted	Achieved
Replay Attack Resistance	Previously captured codes are rejected	Achieved
System Reliability	Stable operation with consistent verification results	Achieved

Observations

- The rolling code mechanism successfully prevented unauthorized access by ensuring that each transmitted signal was unique.
- Replayed signals were rejected because they did not match the expected rolling code within the current counter window.
- The system maintained stable performance with acceptable response time during repeated operations.

Evaluation

The secure system effectively mitigates replay attacks by preventing reuse of captured signals and ensuring message freshness through dynamic code generation. While the system improves resistance to relay attacks, it does not completely eliminate them due to the one-way communication architecture. Overall, the system achieved the intended security objectives while maintaining reliable performance.

7.5 Design Constraint Verification

Table 17. Design Constraint Verification

Constraint	Test Method	Expected Value	Actual Value	% Error	Status
Communication Reliability	Repeated transmission test	$\geq 95\%$ success	~97% success	+2.1%	Met
Response Time	Measured delay	≤ 300 ms	~220 ms	-26.7%	Met
Security	Attack testing	100% rejection in secure mode	100%	0%	Met

7.6 Subsystem Testing

Each subsystem was tested independently:

- **Remote Key:** Verified correct input detection and signal transmission
- **Vehicle Unit:** Verified signal processing and output control
- **Replay Device:** Verified capture, storage, and replay functionality
- **Relay System:** Verified real-time forwarding
- **Secure System:** Verified authentication and access control

All subsystems operated as expected.

7.7 Results Analysis and Discussion

The testing results demonstrate a clear progression:

- **Stage 1:** Functional but insecure system
- **Stage 2:** System vulnerabilities successfully exploited through replay and relay attacks
- **Stage 3:** Rolling code security mechanism effectively mitigates replay attacks and improves overall system security

The prototype achieved all major functional and security objectives. Minor improvements could include:

- Enhancing the robustness of the encoding mechanism
- Increasing communication range stability
- Reducing system latency further

The testing process provided strong evidence that the system meets its design goals. It clearly demonstrates the risks associated with insecure wireless communication and the effectiveness of lightweight rolling code-based security mechanisms in embedded systems.

8. Analysis of the impact of the engineered solution

The proposed system enhances vehicle security by implementing a rolling code authentication mechanism, which significantly reduces the risk of replay attacks and improves resistance to unauthorized access. From a public safety perspective, this solution helps prevent unauthorized vehicle access and potential theft, thereby improving personal security and reducing financial loss. By increasing the reliability of keyless entry systems, the project contributes to user confidence in everyday transportation and supports overall community safety.

From a global and societal perspective, the project demonstrates a low-cost and reproducible security solution that can be applied to older vehicles, especially in regions where advanced security systems are not widely available. This contributes to sustainable development by extending the usability of existing vehicles rather than requiring full system replacement. Additionally, the project raises awareness of cybersecurity in embedded systems and promotes the adoption of secure engineering practices among students and technicians.

From an economic standpoint, the system offers an affordable upgrade that can reduce costs associated with vehicle theft and create opportunities for local businesses in installation and maintenance services. Environmentally, the use of low-power communication modules and efficient electronic components ensures minimal energy consumption and reduced electronic waste, making the solution environmentally responsible.

Table 18. Analysis of the Impact of the Engineered Solution Across Different Contexts

Context	Specific contribution through the project	Level of impact (High, Medium, low)
Public Health, Safety, and Welfare	Improves personal safety by reducing car theft risks. Enhances user confidence in daily transportation. Help prevent break-ins, reduce property loss, and increase safety for families and vehicle owners.	High

Global	Demonstrates a low-cost, reproducible security upgrade suitable for countries with many older vehicles. Support secures mobility worldwide and helps raise awareness of cybersecurity in automotive systems. Contributes to sustainable development by extending the lifespan of existing vehicles through improved security.	Medium
Societal	Increase community safety by reducing theft and unauthorized access. Help educate students/technicians on modern security methods. Encourage adoption of safer digital practices. Promotes trust in smart systems used daily.	High
Environmental	Uses low-power RF modules, reducing energy consumption. System components are lightweight and efficient. Rechargeable or replaceable batteries produce minimal electronic waste. RF emissions are very low and safe.	Low-Medium
Economic	Low-cost design creates opportunities for local businesses (installation, repair, and aftermarket upgrades). Reduces costs associated with car theft. Affordable for wide markets, supports economic competitiveness.	High

9. Conclusion

This project set out to design and evaluate a secure wireless vehicle access system by demonstrating both existing vulnerabilities and effective countermeasures. The main objective was to build a practical prototype that progresses from an insecure baseline system to a secure rolling code–based authentication solution.

9.1 Achievement of Project Objectives

The project objectives were successfully achieved to a high degree:

- **Baseline System (Stage 1):** A functional wireless keyless entry system was implemented. It reliably performed lock and unlock operations with clear feedback using LEDs and a buzzer.
- **Attack Demonstration (Stage 2):** Both replay and relay attacks were successfully demonstrated. The system allowed unauthorized access when exposed to these attacks, confirming real-world vulnerabilities.
- **Secure System (Stage 3):** A rolling code–based authentication mechanism was implemented. The system effectively prevented replay attacks and improved resistance to unauthorized access by validating dynamically generated codes.

Overall, the project achieved its goal of demonstrating the full security lifecycle: vulnerability → exploitation → protection.

9.2 Strengths of the Solution

The proposed solution has several key strengths:

- **Clear Demonstration of Security Concepts:** The three-stage design provides a structured and educational approach to understanding wireless security.
- **Low-Cost and Practical Implementation:** The system was built using affordable and widely available components, making it scalable and reproducible.
- **Effective Attack Simulation:** Replay and relay attacks were implemented successfully, providing realistic validation of system weaknesses.

- **Efficient Security Mechanism:** The rolling code mechanism effectively mitigates replay attacks while maintaining low computational complexity.
- **Modular Design:** The system can be easily extended or modified for future improvements.

9.3 Limitations of the System

Despite its success, the system has some limitations:

- **Limited Communication Range:** RF communication performance may vary depending on environmental conditions.
- **Lightweight Security Implementation:** While effective against replay attacks, the rolling code mechanism may not withstand advanced attacks without stronger cryptographic techniques.
- **Prototype-Level Design:** The system is not yet optimized for commercial deployment or large-scale manufacturing.
- **Synchronization Dependency:** The system relies on counter synchronization, which may be affected if multiple transmissions are missed.

9.4 Key Contributions and Novel Aspects

This project contributes to both academic learning and practical engineering:

- **Integration of Attack and Defense in One System:** The project demonstrates both vulnerabilities and security solutions within a single platform.
- **Three-Stage Security Framework:** The structured progression (baseline → attack → secure system) provides a clear methodology for analyzing wireless security.
- **Real-Time Attack Demonstration:** Implementation of replay and relay attacks using embedded systems adds significant practical value.
- **Low-Cost Secure System Design:** The project shows that effective security can be achieved using lightweight and affordable embedded solutions.

9.5 Final Remarks

In conclusion, the project successfully demonstrates how insecure wireless systems can be exploited and how appropriate security mechanisms, such as rolling code authentication, can mitigate these risks. It highlights the importance of integrating security into system design from the early stages. The results provide valuable insights into wireless communication security and establish a strong foundation for future development in secure embedded systems and automotive technologies.

10. Future Work

While the proposed system successfully demonstrates the full security lifecycle of wireless vehicle access systems, there are several areas where the design can be further improved and extended. Future work can enhance the system's performance, security, scalability, and real-world applicability.

10.1 Security Enhancements

- **Stronger Cryptographic Techniques:** The current rolling code mechanism can be enhanced by integrating stronger cryptographic algorithms (e.g., AES-based encoding) to improve resistance against advanced attacks and cryptanalysis.
- **Enhanced Rolling Code Mechanism:** Improving the transformation function and increasing key complexity can strengthen the existing rolling code implementation and reduce predictability.
- **Anti-Relay Techniques:** Additional countermeasures such as time-based validation or distance-bounding techniques can be implemented to better detect and mitigate relay attacks.

10.2 Hardware Improvements

- **Custom PCB Design:** Transitioning from breadboard prototyping to a printed circuit board (PCB) would improve system reliability, compactness, and suitability for real-world deployment.
- **Improved Antenna Design:** Optimizing antenna configuration can enhance communication range and reduce interference.
- **Power Optimization:** Implementing low-power modes and energy-efficient components would make the system more suitable for battery-operated applications.

10.3 Software and System Enhancements

- **Error Detection and Correction:** Adding mechanisms such as checksums or CRC can improve communication reliability.
- **Enhanced Communication Protocol:** Developing a more structured communication protocol with validation and integrity checks would improve system robustness.
- **Data Logging and Monitoring:** Integrating logging features to record system activity and attacks would support further analysis and debugging.

10.4 User Interface and Usability

- **Mobile Application Integration:** Developing a smartphone application for system control and monitoring would improve usability and expand functionality.
- **Graphical User Interface (GUI):** A user-friendly interface could provide real-time system status, alerts, and control options.

10.5 System Scalability and Integration

- **Integration with Smart Systems:** The system can be extended to integrate with IoT platforms or smart vehicle systems.
- **Multi-User Support:** Future versions can support multiple keys with unique authentication credentials.

- **Cloud Connectivity:** Adding cloud-based monitoring and control would allow remote access and system management.

10.6 Additional Resources Required

To implement these improvements, the following resources may be required:

- Advanced cryptographic libraries and security frameworks
- PCB design tools and fabrication services
- Additional hardware modules (e.g., secure elements, sensors)
- Software development tools for mobile and cloud integration
- Extended testing equipment for performance and security validation

Future work can significantly enhance the system by improving its security, reliability, and usability. By addressing current limitations and integrating advanced technologies, the proposed solution can evolve from a prototype into a robust and scalable real-world system.

11. Student reflections

Student Amna:

Throughout the development of this project, I gained significant technical, analytical, and leadership experience that contributed greatly to my growth as an engineering student. As the team leader, I was responsible for guiding the overall direction of the project, coordinating tasks among team members, organizing meetings and progress discussions, and ensuring that project objectives and deadlines were consistently met. Initially, the project appeared highly complex due to the integration of cybersecurity concepts, embedded systems, wireless communication, and hardware implementation. However, through effective planning, continuous communication with our supervisor, and structured teamwork, the project became more manageable and technically clearer throughout each stage of development.

A major part of my contribution involved leading the research and analysis of vulnerabilities in existing smart key fob systems, particularly focusing on replay

attacks and relay attacks that exploit insecure wireless communication protocols. This required extensive investigation into RF communication methods, authentication weaknesses, and common attack vectors affecting traditional keyless entry systems. Through this process, I developed a deeper understanding of practical cybersecurity risks within embedded and IoT based systems and learned the importance of integrating security considerations during the early design stages rather than treating them as secondary features. The analytical work and security research I conducted played a major role in shaping the architecture and logic of our prototype system. In addition, I supervised the planning of hardware selection, RF module integration, system architecture design, and the organization of technical documentation and report structure. Managing both technical and documentation related responsibilities strengthened my ability to balance engineering implementation with professional reporting standards. From a professional perspective, leading this project significantly improved my leadership, communication, and problem solving abilities. It taught me the importance of decision making under pressure, maintaining team coordination, and ensuring accountability throughout the project lifecycle. Furthermore, this experience strengthened my confidence in conducting technical research, interpreting hardware data sheets, and translating theoretical cybersecurity concepts into a practical engineering solution. Overall, this project enhanced both my technical expertise and my ability to lead and manage a multidisciplinary engineering project in a professional and organized manner. From a DEIA perspective, our team maintained a respectful environment with open communication. Everyone contributed fairly, and feedback from the supervisor was provided equally to all members.

Student Aldana

During Stages 1 and 2, I gained extensive hands on engineering experience through leading the hardware implementation and attack simulation aspects of the project. I was heavily involved in assembling circuits, interpreting technical datasheets, configuring microcontrollers, testing RF communication behavior, and

troubleshooting hardware related issues throughout the entire development process. I took primary responsibility for implementing and validating the attack mechanisms within the prototype system, including analyzing vulnerabilities, testing communication responses, and ensuring that the simulated attack scenarios operated accurately and reliably. This role required continuous testing, debugging, and refinement of both the hardware and system behavior under different operating conditions.

The project significantly strengthened my practical engineering skills and reinforced my confidence in working with embedded systems and real hardware environments. I developed a deeper understanding of circuit integration, signal behavior, communication reliability, and system level troubleshooting. Being trusted with major technical responsibilities gave me a strong sense of ownership over the project and pushed me to think critically and solve problems independently under time and implementation constraints. My contribution played a major role in transforming the project from an initial concept into a functioning prototype system.

Beyond the technical aspects, this experience also improved my leadership and communication skills. I became more confident when contributing during technical discussions, explaining engineering decisions, and collaborating with the team to resolve implementation challenges

Student Latifa

During Stage 1, I focused primarily on assisting with the integration between the hardware and the software components of the system. This allowed me to develop a solid understanding of how microcontroller functions, firmware logic, and communication protocols come together to form a working prototype. I gained valuable experience in testing modules, observing system performance, and verifying that the design met the intended requirements. This strengthened my interest in embedded system development and deepened my understanding of secure wireless design. These lessons will continue to guide me as we move into the next stages involving attack simulations, and defense mechanisms.

Student Muneera

In Stage 1, I developed strong technical and professional skills through my active involvement in the digital and presentation aspects of the project. I contributed

significantly to the creation and organization of the project's visual content, including designing presentation materials, preparing system visuals, and assisting in the development of the pitching video used to present the project concept and objectives. Working on the digital side of the project strengthened my ability to communicate technical ideas in a clear, structured, and visually engaging manner, which is highly important when presenting engineering solutions to both technical and non technical audiences.

In addition, I played an active role in the after market and user focused side of the project by contributing to the visual identity and presentation of the prototype concept. This included helping develop the overall appearance, presentation style, and promotional aspects of the system to make the project more understandable and appealing. Through these responsibilities, I improved my creativity, teamwork, and communication skills while learning the importance of how engineering solutions are presented and perceived by users and stakeholders. This experience increased my confidence in collaborating within a team environment, participating in technical discussions, and supporting the project beyond only the technical implementation. It also taught me the importance of combining technical understanding with effective presentation, visualization, and communication skills to create a more complete and professional engineering project.

References

- [1] A. Francillon, B. Danev, and S. Capkun, “Relay Attacks on Passive Keyless Entry and Start Systems in Modern Cars,” Network and Distributed System Security Symposium (NDSS), 2011.
- [2] W. Verdult, F. D. Garcia, and J. Balasch, “Gone in 360 Seconds: Hijacking with Hitag2,” USENIX Security Symposium, 2012.
- [3] F. D. Garcia et al., “On Vehicular Security for RKE and Cryptographic Algorithms: A Survey,” ACM Computing Surveys, vol. 53, no. 5, 2020.
- [4] T. Van Capelleveen, “Security Analysis of Aftermarket Remote Keyless Entry Systems for Consumer Vehicles,” Master’s Thesis, Radboud University, 2020.
- [5] S. Brands and D. Chaum, “Distance-Bounding Protocols,” Advances in Cryptology – EUROCRYPT ’93, Springer, 1993.
- [6] K. Rasmussen and S. Capkun, “Realization of RF Distance Bounding,” USENIX Security Symposium, 2010.
- [7] K. H. Lee et al., “Design of Secure Challenge-Response Protocols for Wireless Car Keys,” IEEE Access, vol. 8, pp. 11094–11105, 2020.
- [8] M. Al-Dosari, H. Al-Khalifa, and F. A. Qureshi, “Key is in the Air: Hacking Remote Keyless Entry Systems,” Hamad Bin Khalifa University / Qatar University Technical Report, 2022.
- [9] NXP Semiconductors, “Rolling Code Security in Remote Keyless Entry Systems,” Application Note AN1234, 2021.
- [10] Texas Instruments, “Implementation of Secure RKE Using AES Encryption,” Technical White Paper, 2020.

- [12] A. Kleanthous, T. Chothia, and A. Malacaria, "Lightweight Authentication Protocols for Keyless Entry Systems," in Proceedings of the IEEE International Conference on Smart Embedded Systems (SES), 2019.
- [13] H. Yildiz, E. Karapistici, and M. Güneş, "Vulnerabilities in Aftermarket RF Remote Control Systems Used in Vehicles," Journal of Information Security and Applications, vol. 54, 2021.
- [14] E. Fernandes, B. Crispo, and N. Borisov, "KeyfobGuard: Behavioral Authentication for Remote Keyless Entry Systems," in Proceedings of the ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec), 2020.
- [15] M. A. Lighari, S. Yu, and R. Khan, "A Secure and Efficient Rolling-Code Generation Mechanism for Wireless Vehicles," IEEE Access, vol. 9, pp. 147212–147225, 2021.
- [16] J. Hines, M. Green, and A. Clark, "Security Analysis of Automotive Telematics and Keyless Entry Systems," IEEE Transactions on Vehicular Technology, vol. 70, no. 4, pp. 3205–3218, 2021.
- [17] R. Spreitzer, V. Moonsamy, T
- [18] I. K. Kim and J. H. Cheon, "Security Analysis of Rolling Code Algorithms in Car Remote Key Systems," IEEE Transactions on Information Forensics and Security, vol. 15, pp. 1234–1247, 2020.
- [19] M. Wolf and T. Gendrullis, "Design, Implementation, and Evaluation of a Vehicular Immobilizer System," Information Security Conference (ISC), Springer, 2012.
- [20] A. Singh, B. Shrestha, and K. Zeng, "A Practical Relay Attack on Passive Keyless Entry Systems," IEEE Conference on Communications and Network Security (CNS), 2019.

[21] J. Han, X. Li, and P. Schartner, “Enhanced Distance-Bounding Protocol for Secure Automotive Access Systems,” *IEEE Access*, vol. 10, pp. 12832–12845, 2022.

[22] V. Draskovic and P. Schartner, “Secure Proximity Verification for Passive Keyless Entry Systems,” *IFIP SEC*, Springer, 2021.

[23] A. Shwartz, A. Segev, O. Shmueli, and A. Wool, “Dramatically Improving the Security of Automotive Remote Keyless Entry Systems,” *IEEE Transactions on Dependable and Secure Computing*, 2022.

Appendix A – Culmination of Design Experiences

Throughout the Computer Engineering program, various courses directly supported the development of our project. The literature review stage was strengthened by courses such as CMPS 380 which introduced cybersecurity principles, vulnerability, and attack types. High-level design skills were obtained from courses like CMPE 462 and CMPE 355, where we learned system architecture, component selection, and design planning.

Implementation abilities were developed through hands-on courses such as ELEC 201, CMPE 370, CMPE 261, and CMPE 462, which trained us in microcontroller interfacing and circuit wiring. Finally, our coding comes from CMPE 462 Arduino-based programming throughout the curriculum. Together, these courses built the knowledge and skills that enabled us to design, implement, and test our senior design project effectively.

